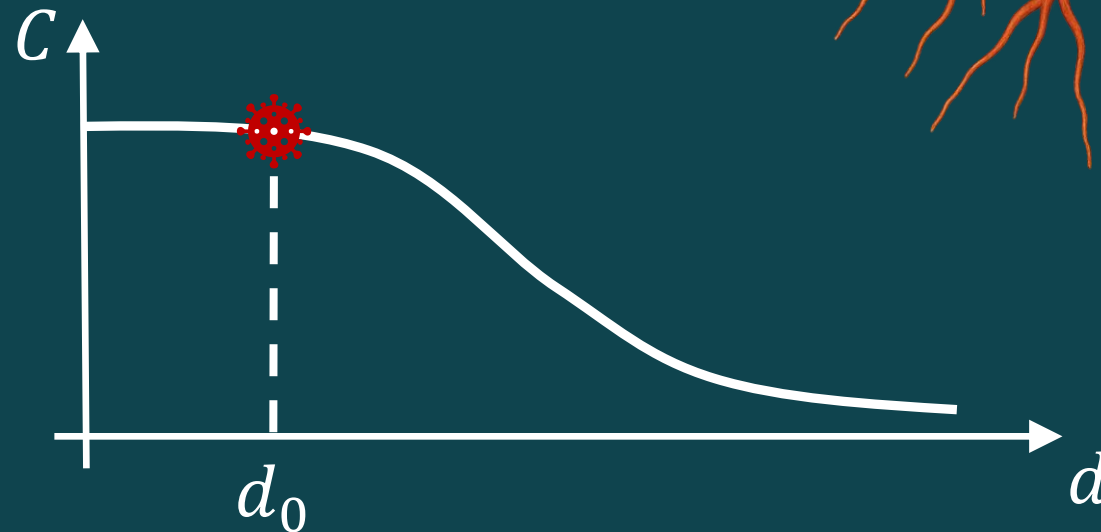
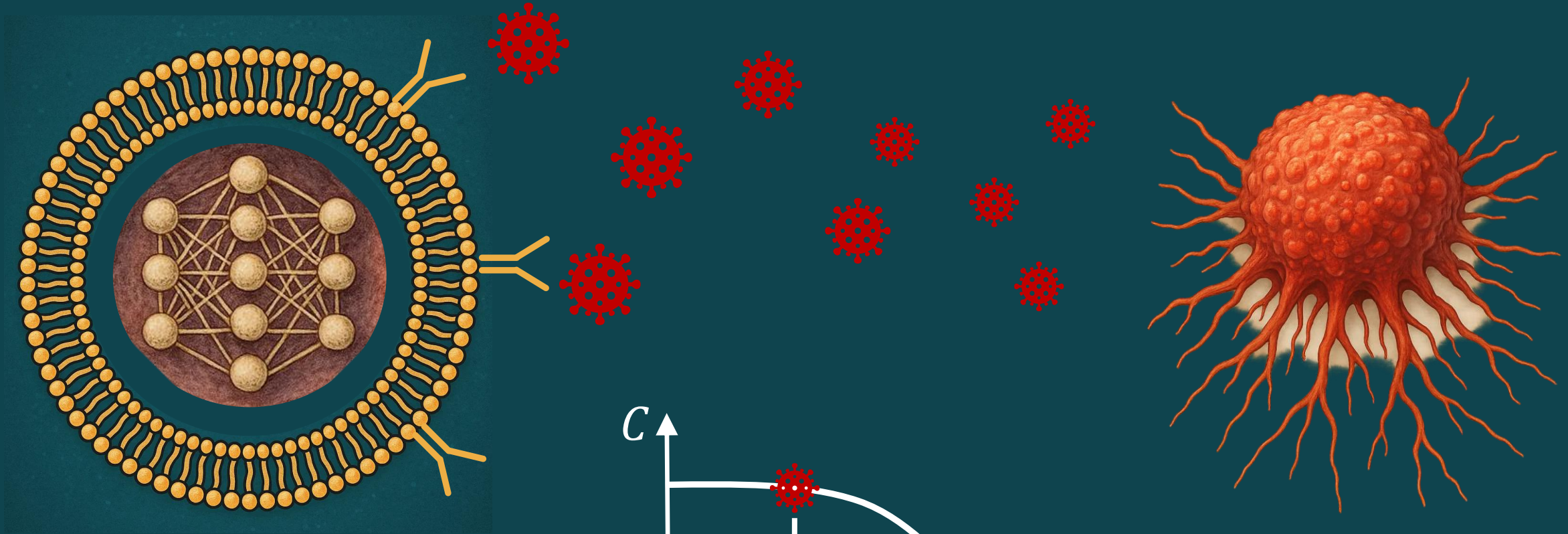
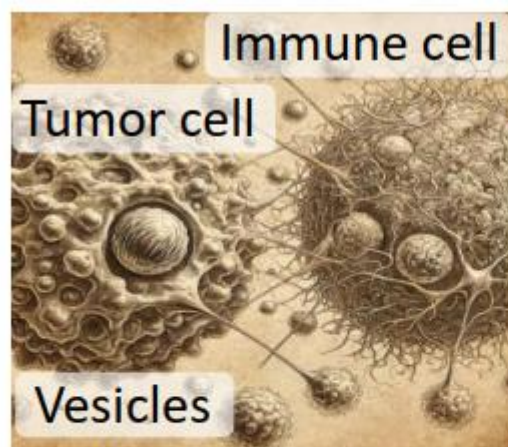


Associating Metabolic reactions with NN coefficients

Bio-Building Carriers

Cancer Cell





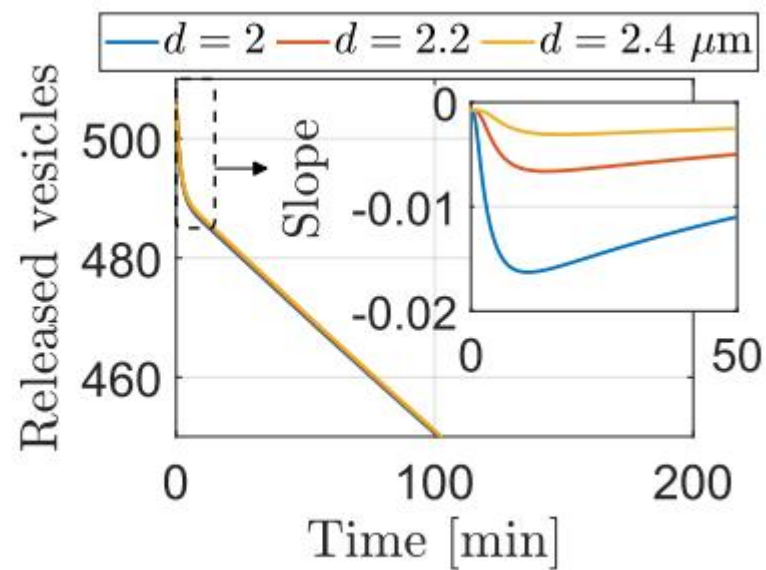
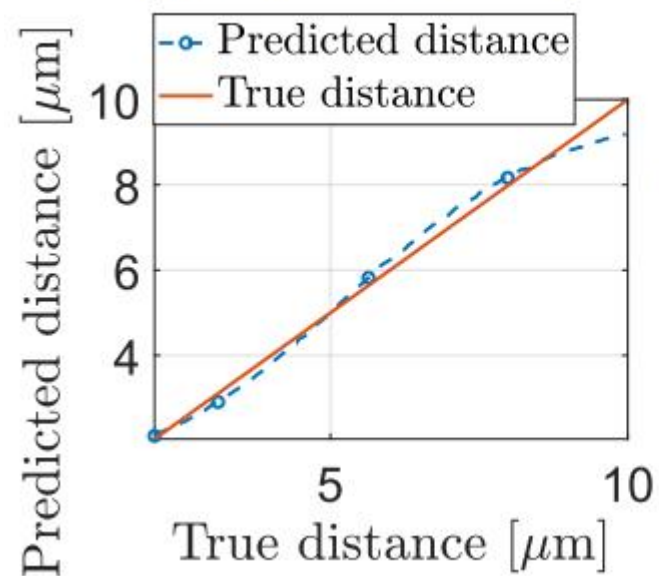
(a) Dataset

Feature
Extraction

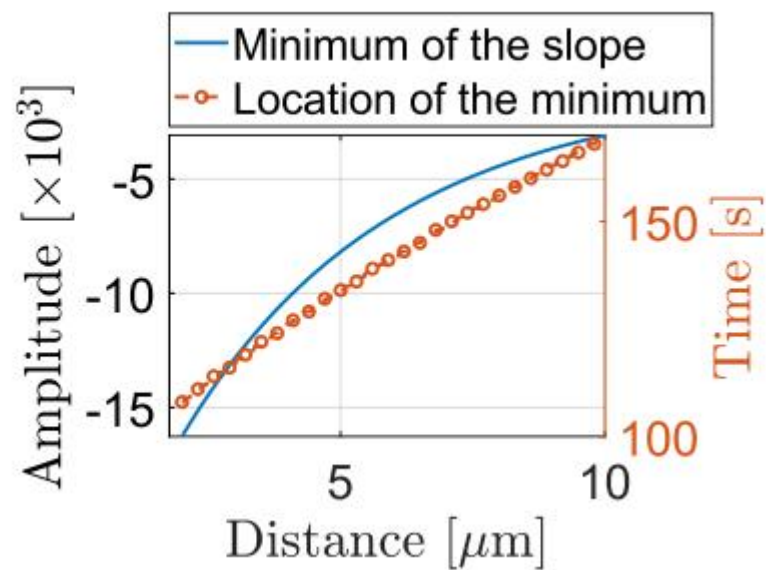
Neural
Network



(c) Estimated distance

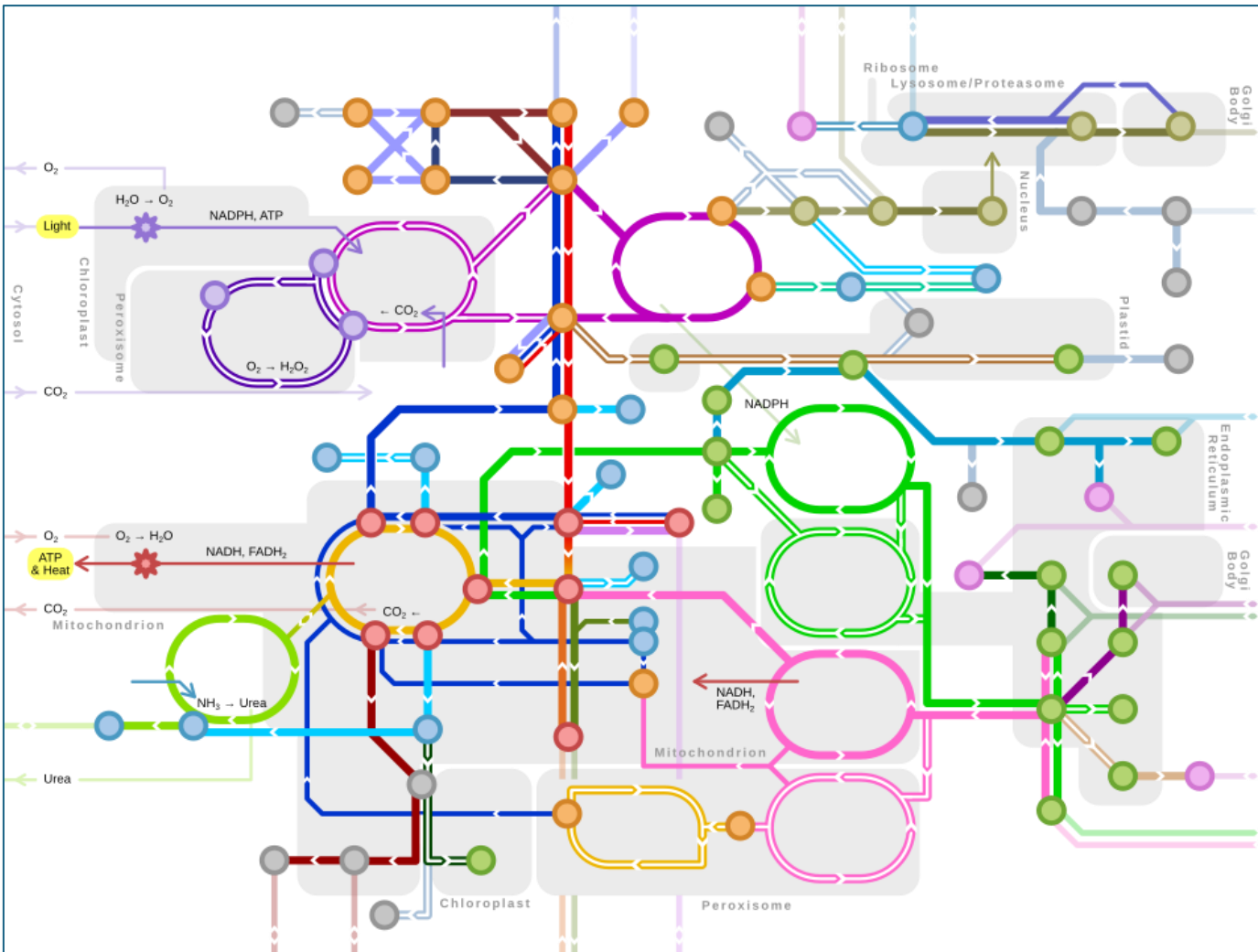


(b) Features

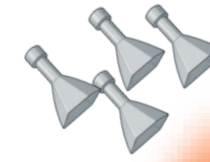


Concept

Metabolic Network



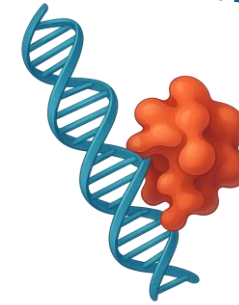
Antenna Array



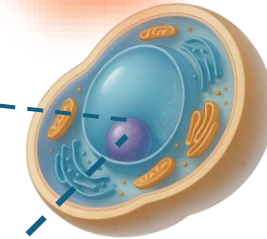
Terahertz Waveforms



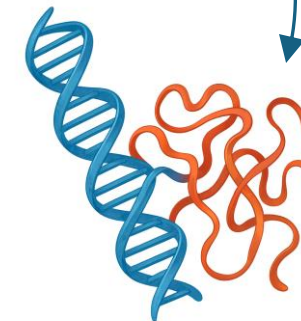
Folded Enzyme



Yeast Cell



Unfolded Enzyme



- Enzyme unfolding probability

$$p_U = 1 - \frac{1}{1 + e^{\left(\frac{\Delta E - \langle E_{tot} \rangle_{ss}}{k_b T}\right)}}$$

$\langle E_{tot} \rangle_{ss}$ Total energy of the enzyme

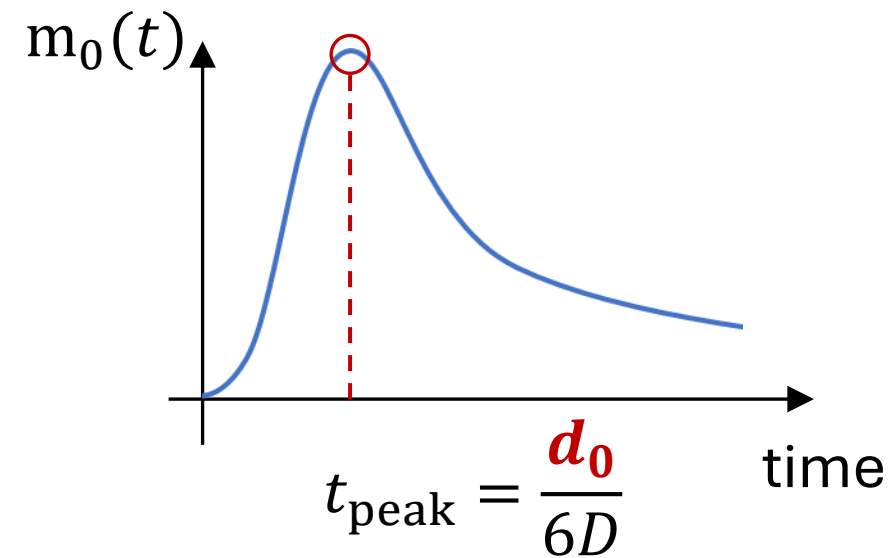
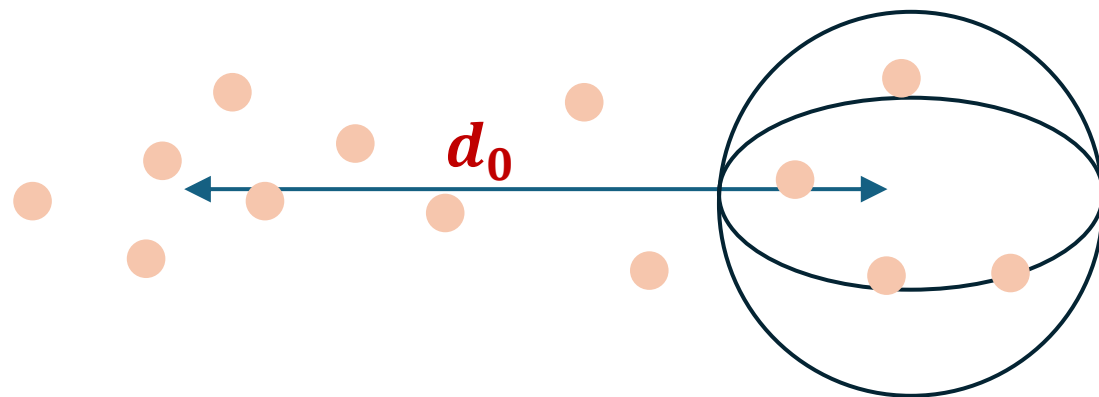
ΔE Enzyme-free energy

k_b Boltzmann constant

T Temperature of the Cell

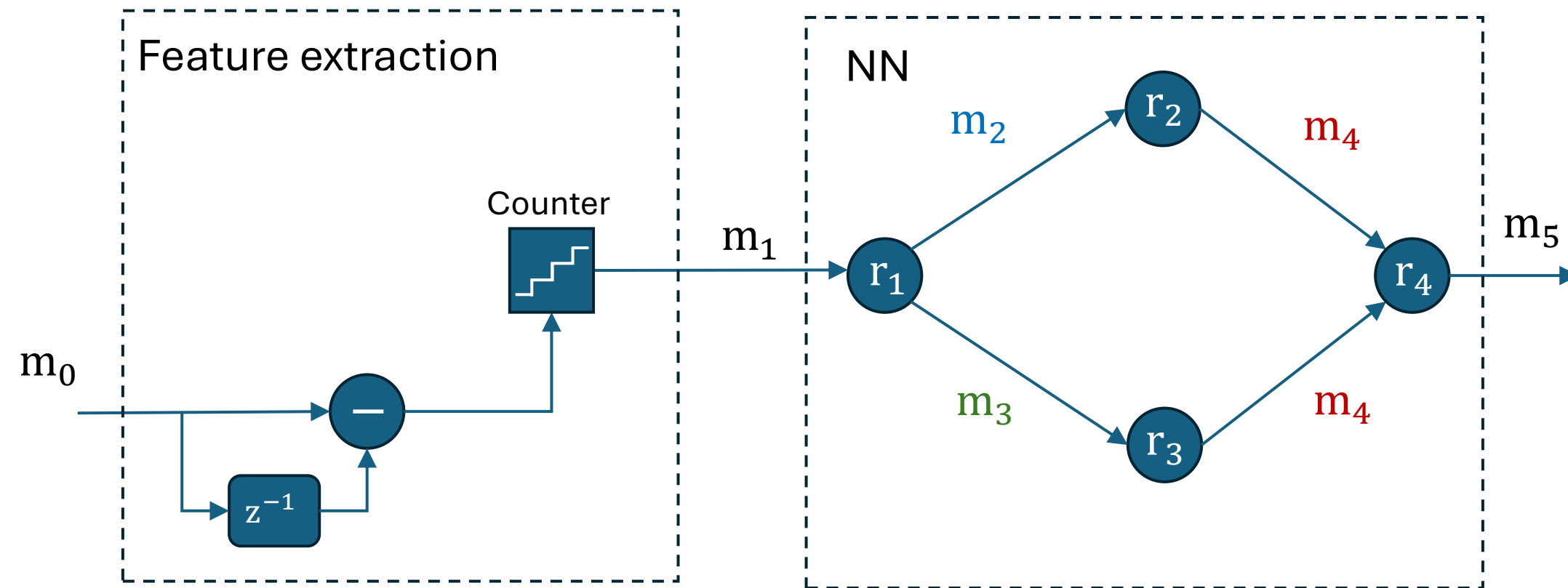
Point Transmitter

Receiver



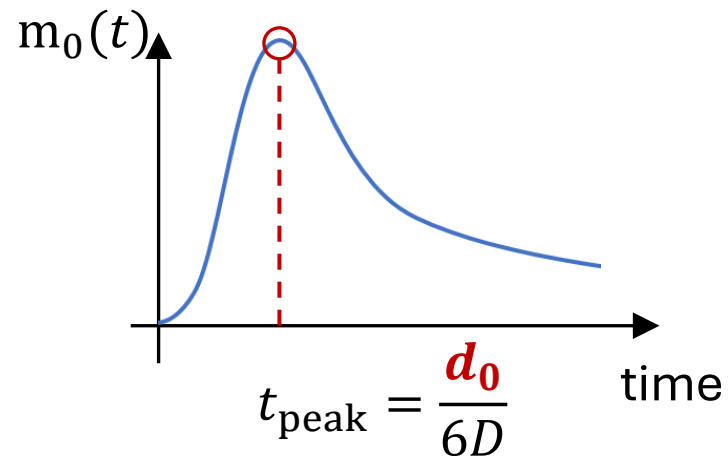
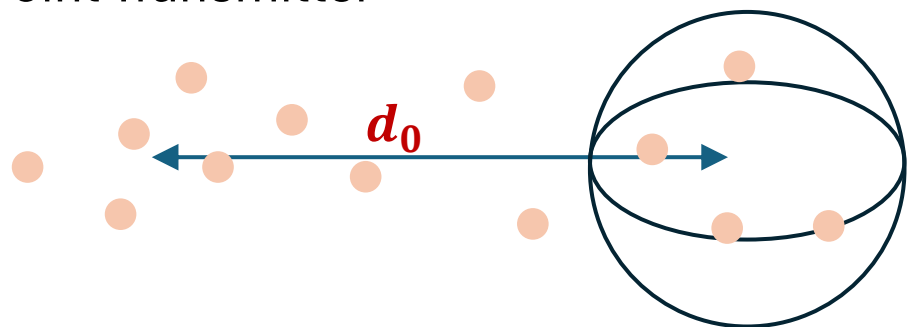
Feature extraction

NN



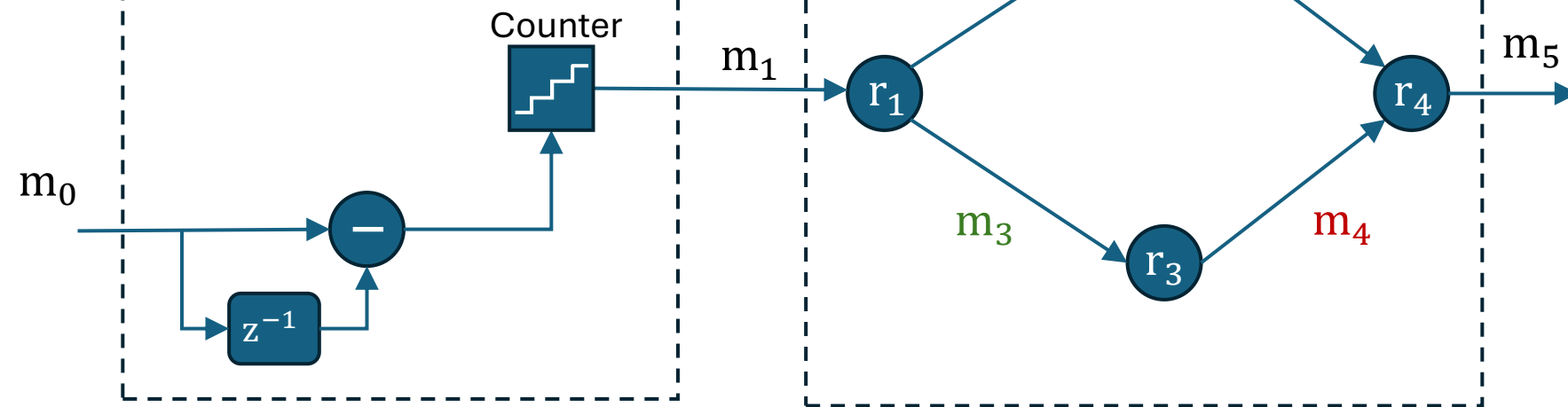
Point Transmitter

Receiver



Feature extraction

NN

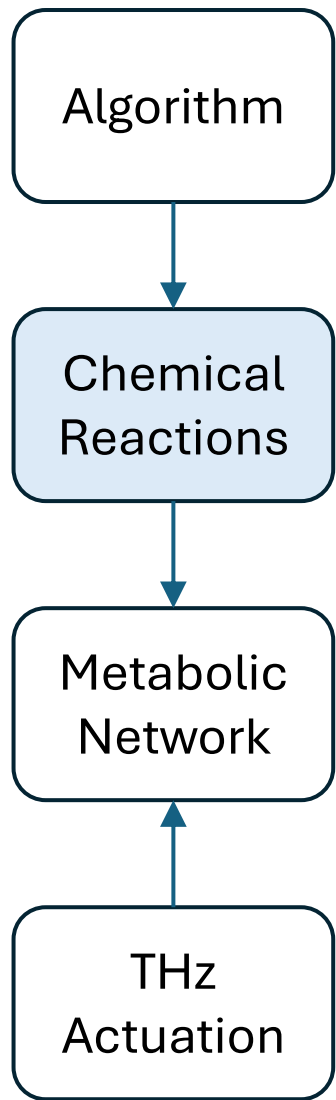
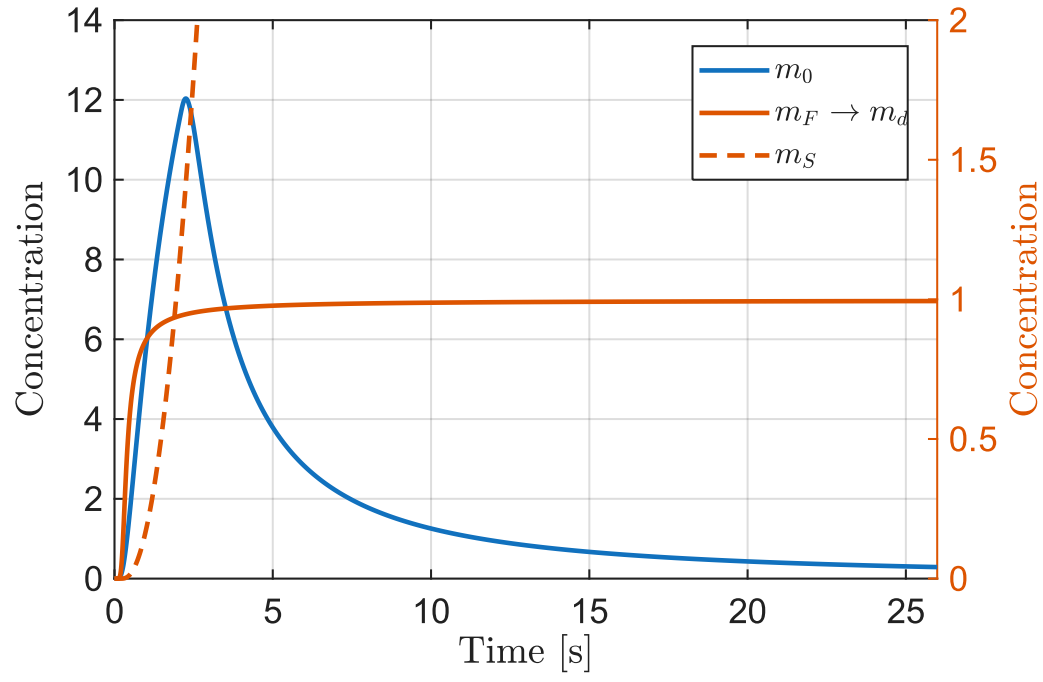
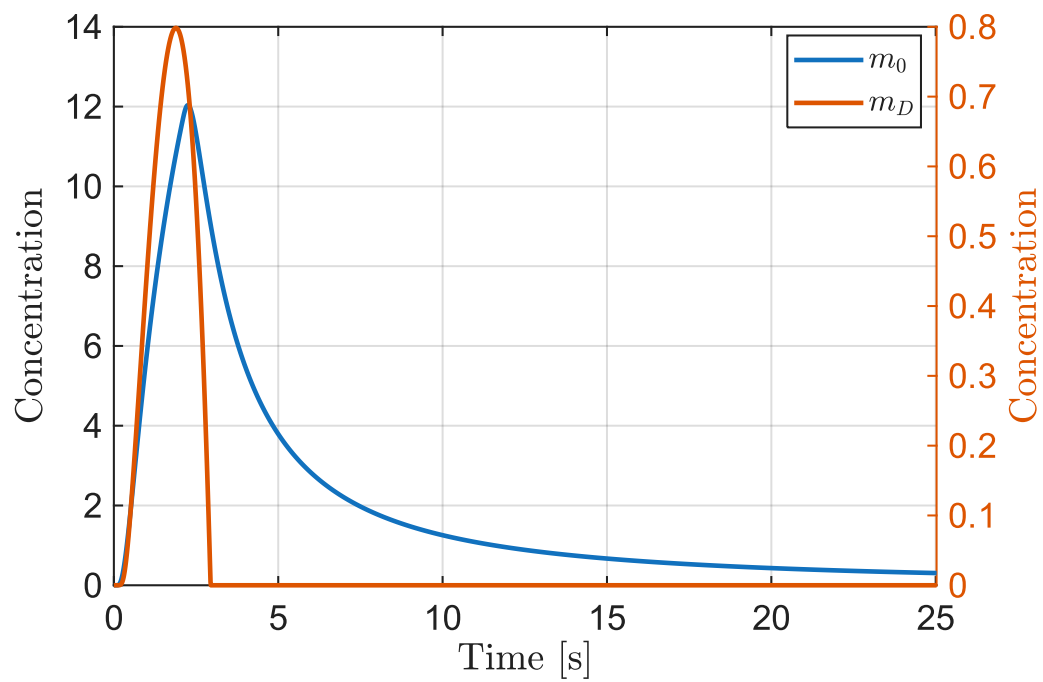
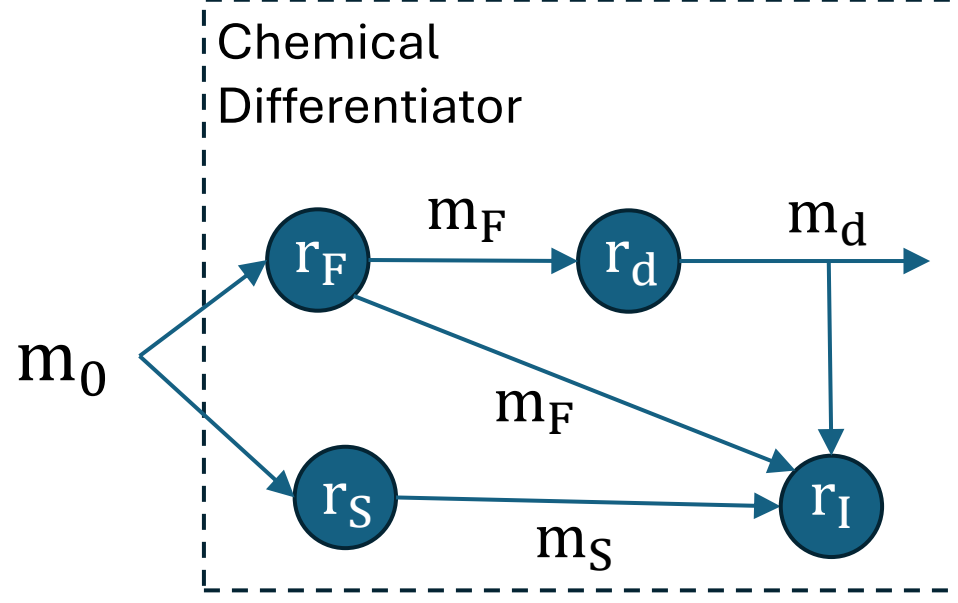
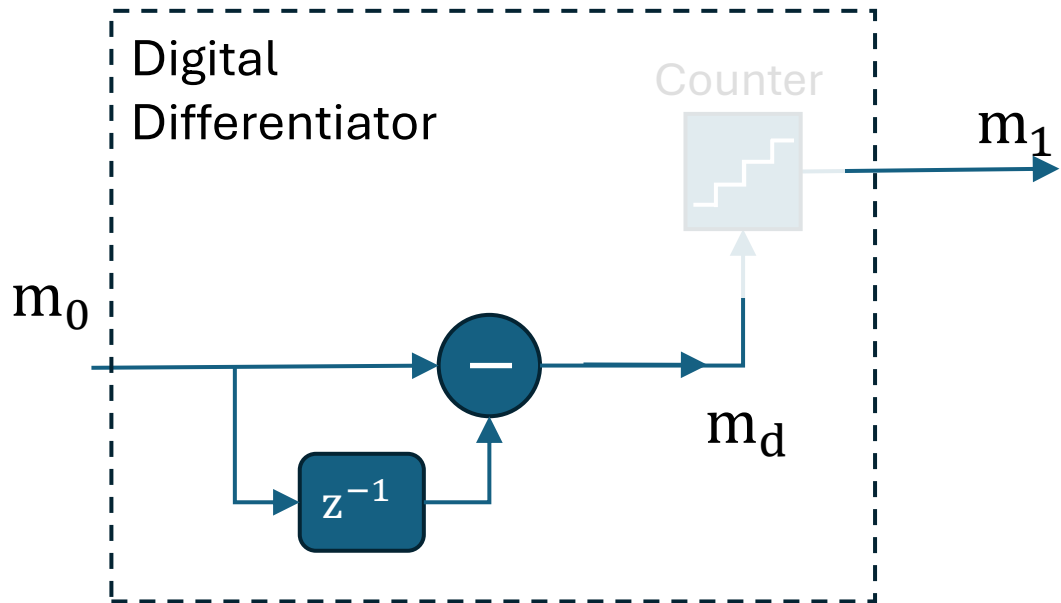


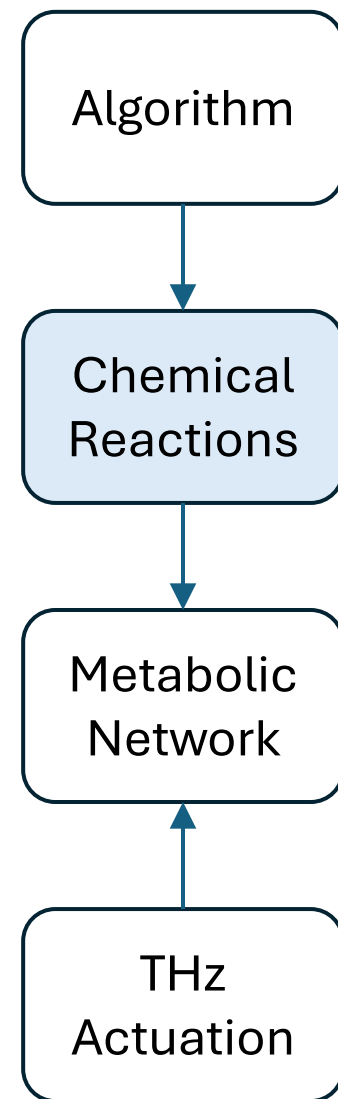
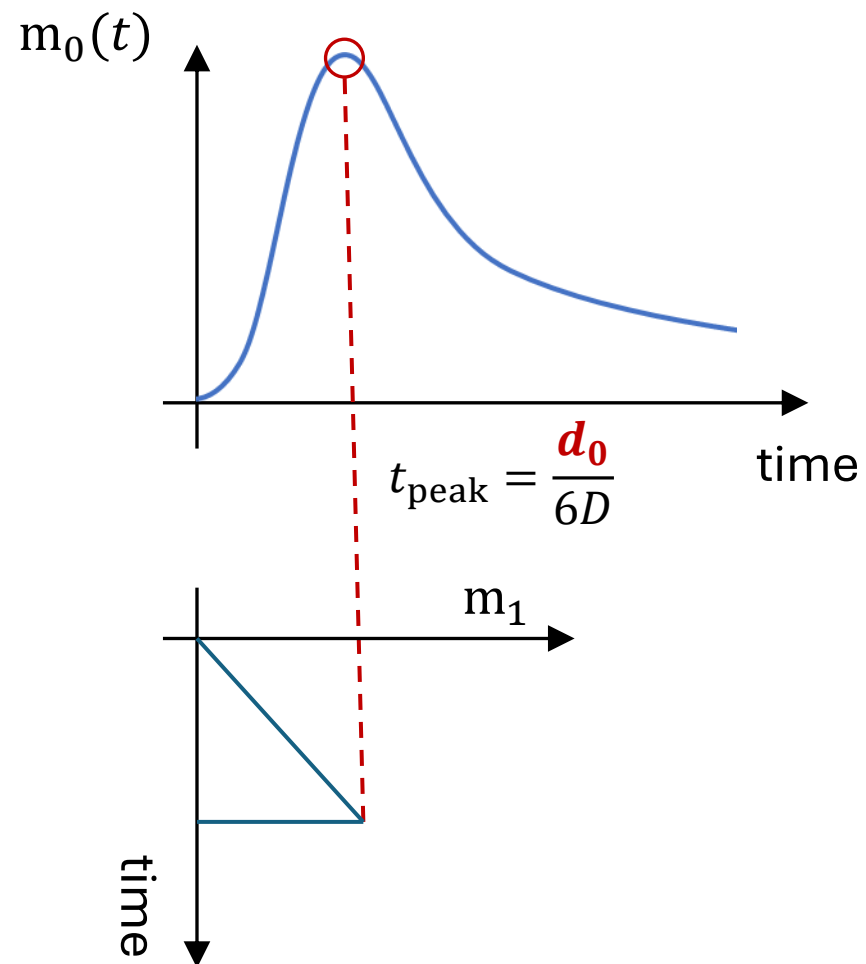
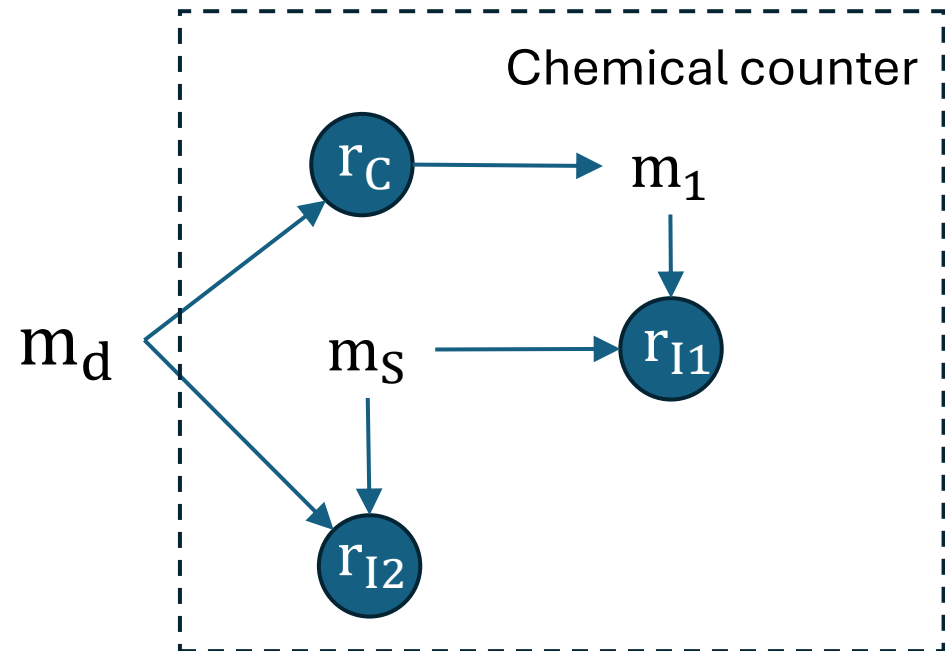
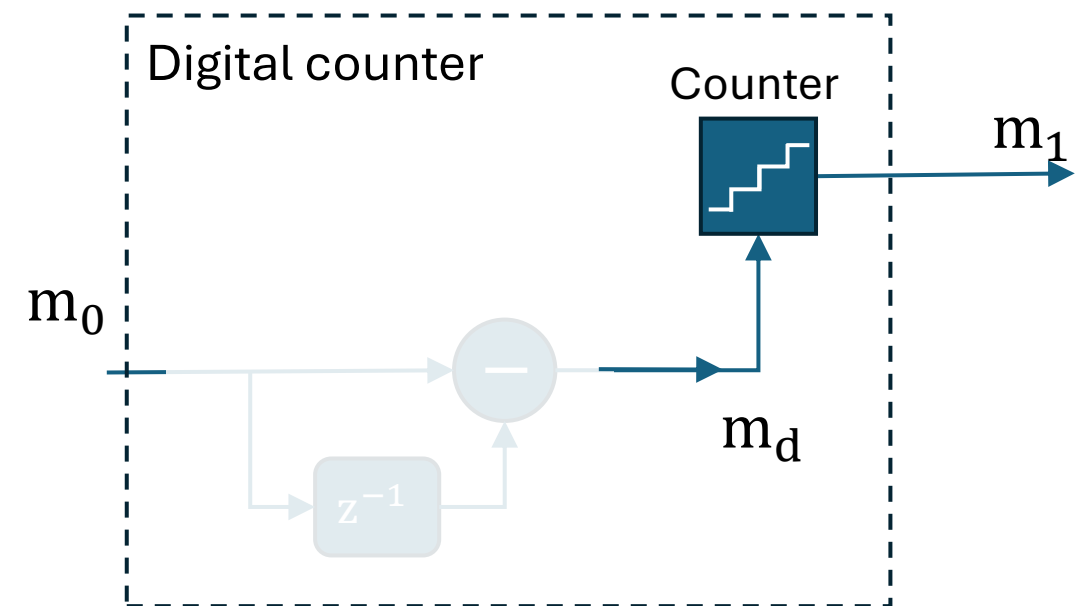
Algorithm

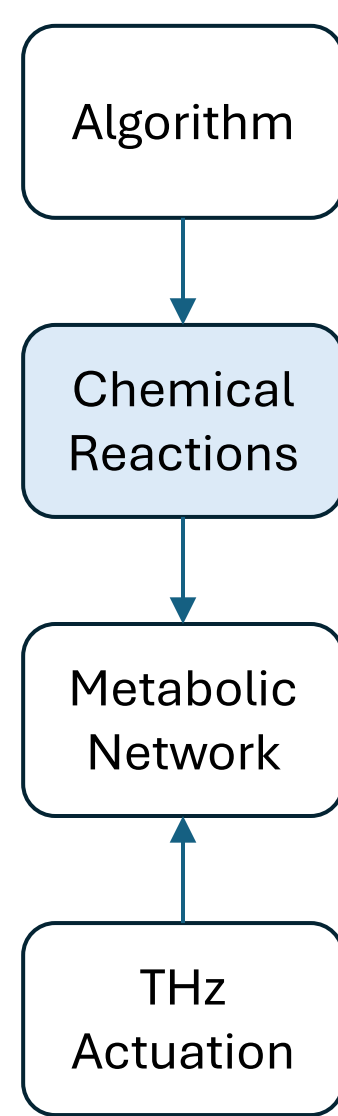
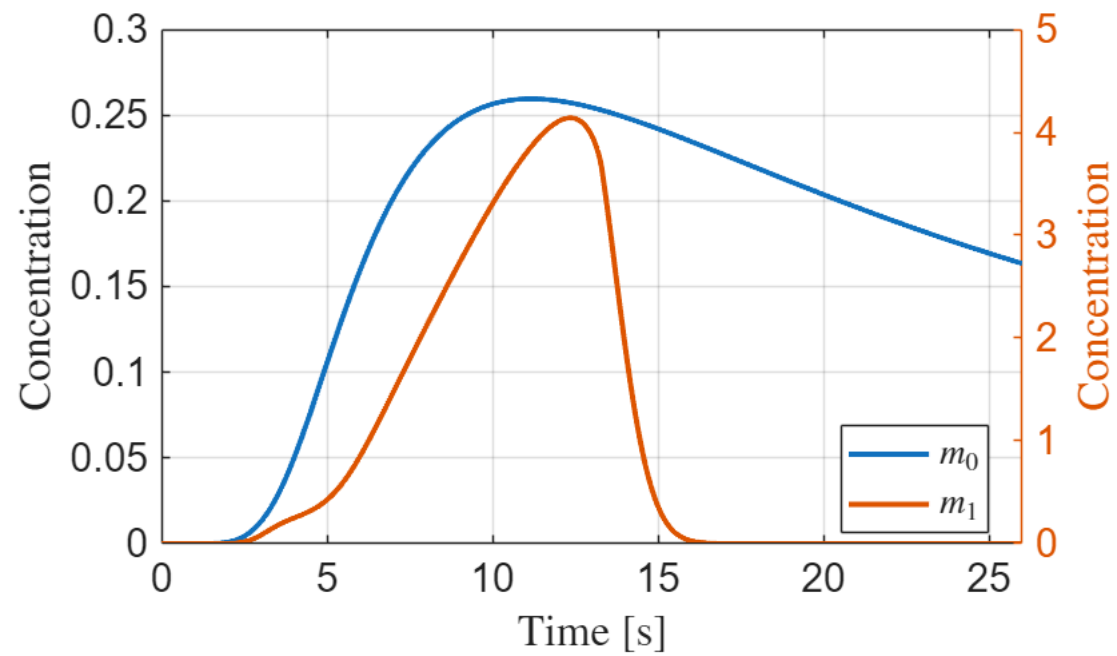
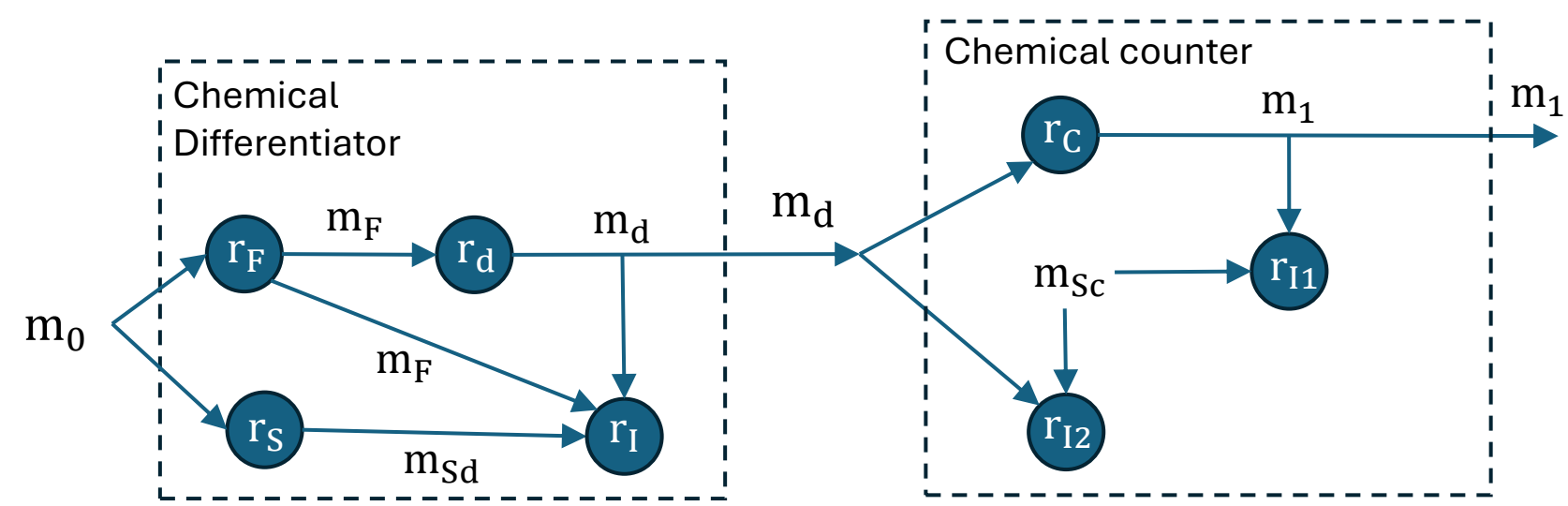
Chemical
Reactions

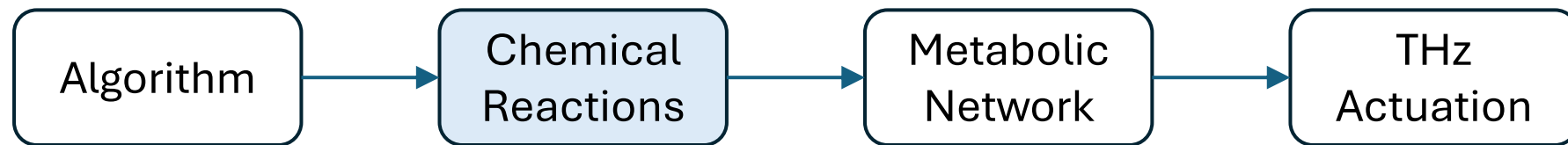
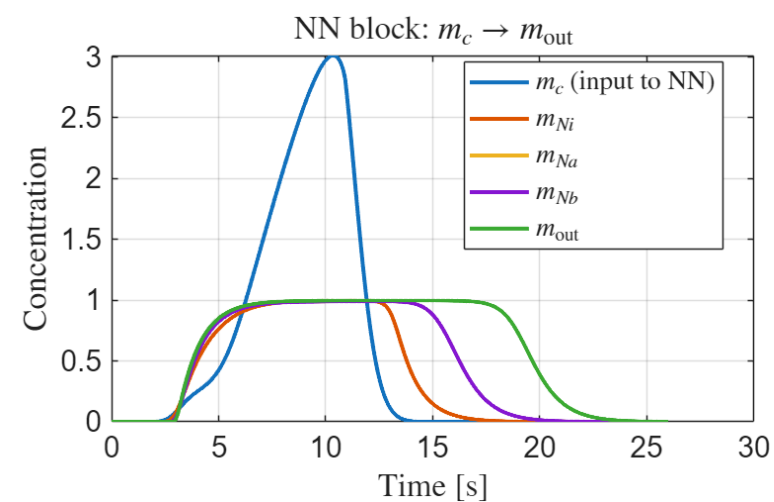
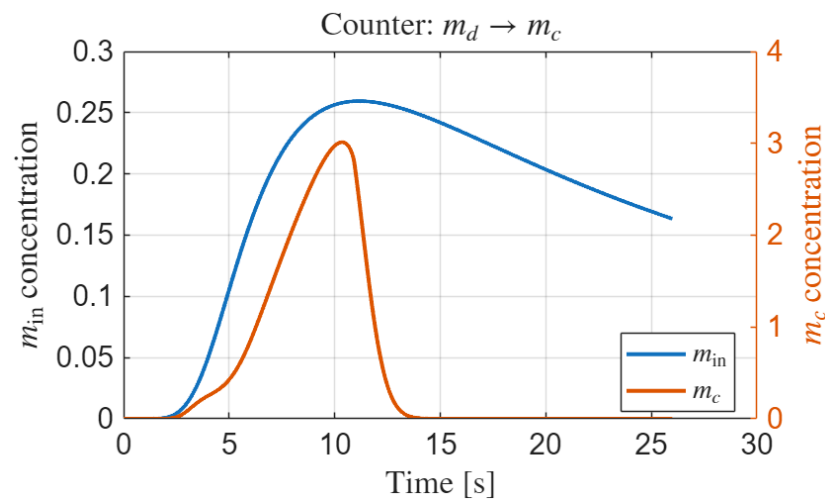
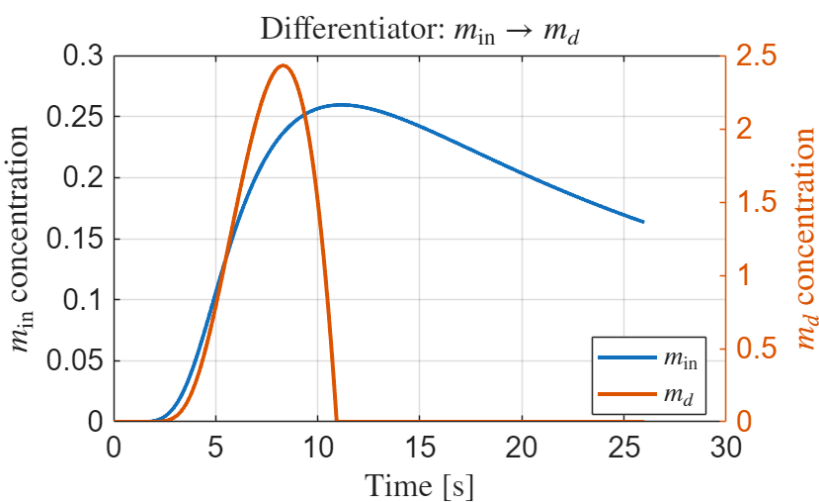
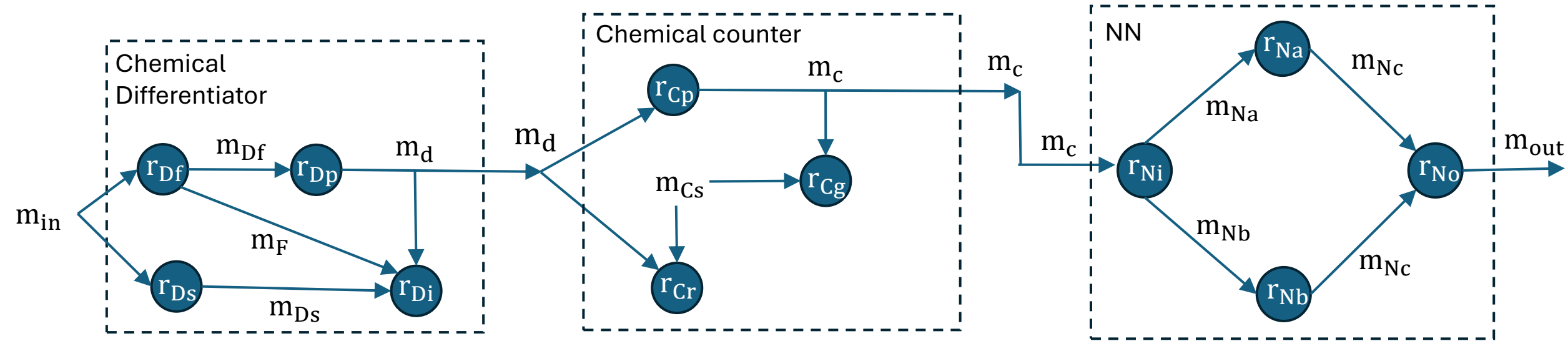
Metabolic
Network

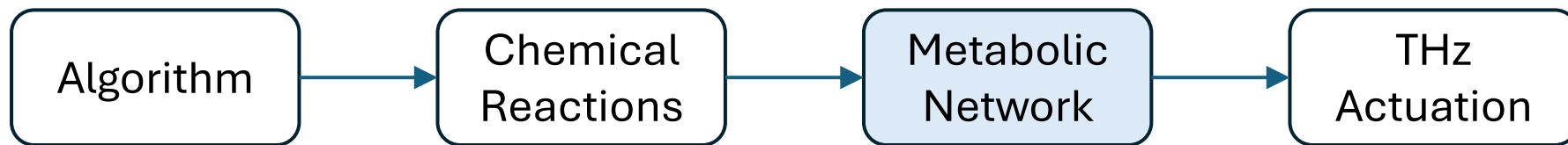
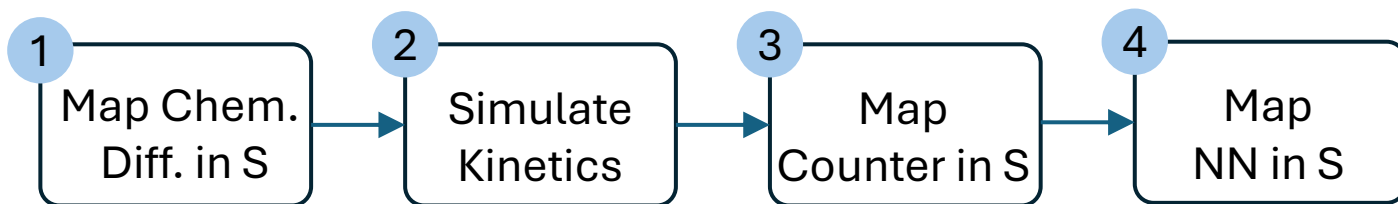
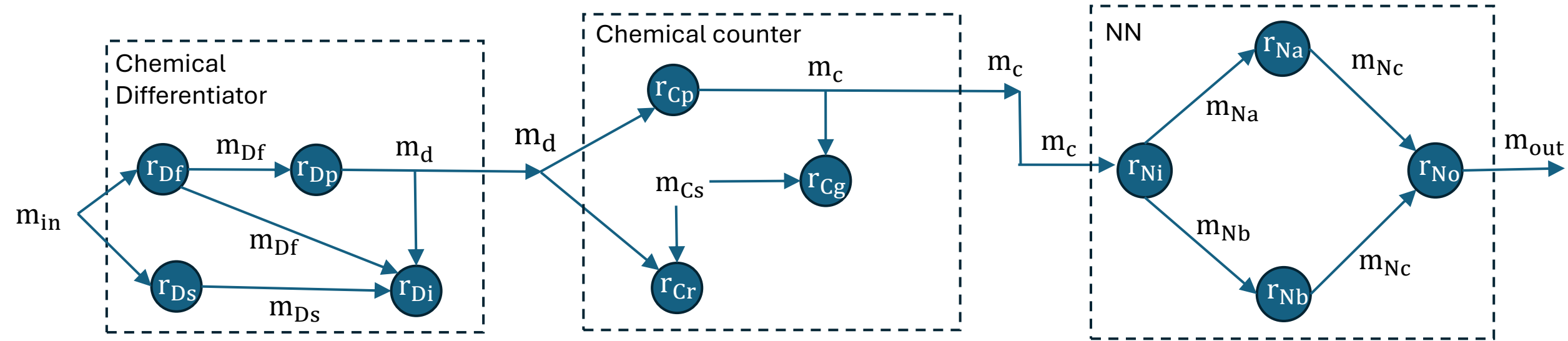
THz
Actuation

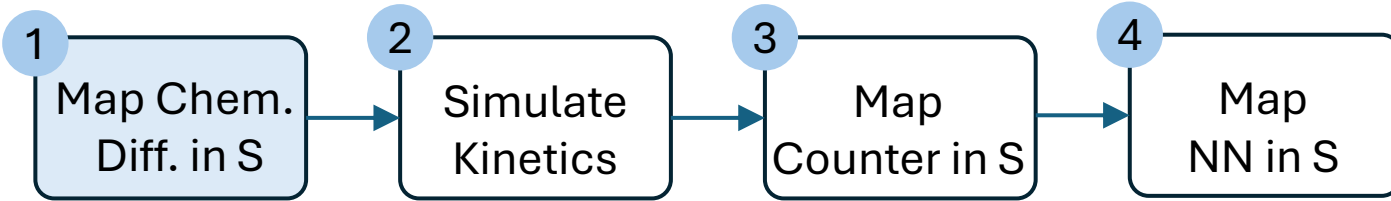
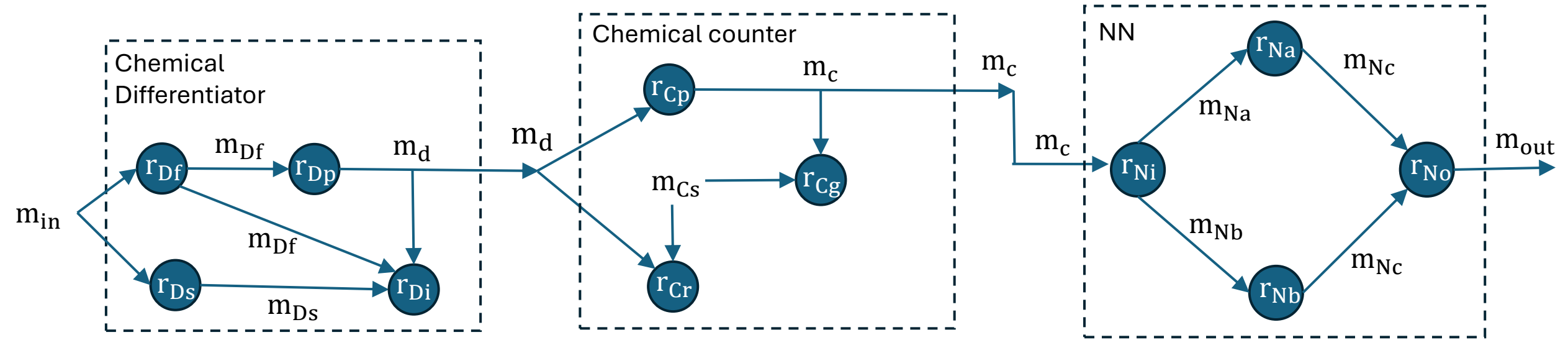












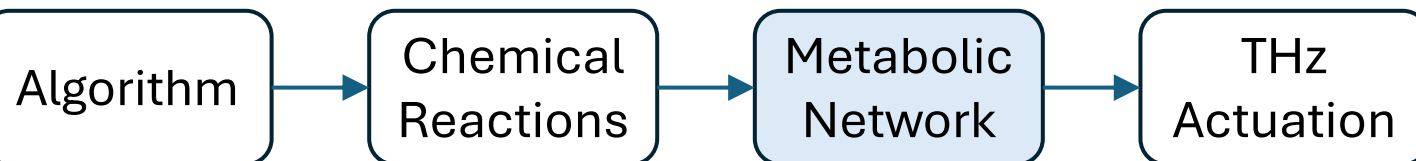
Total candidates (4-reaction topology): 22

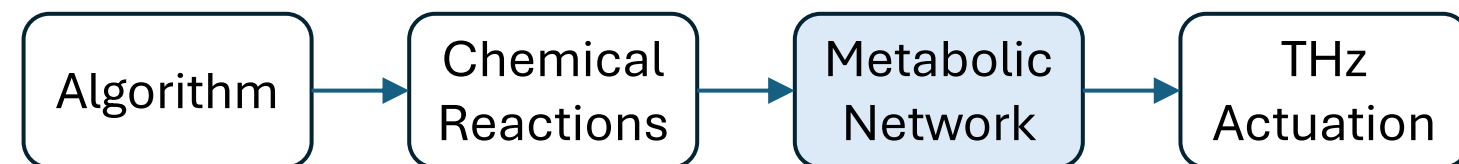
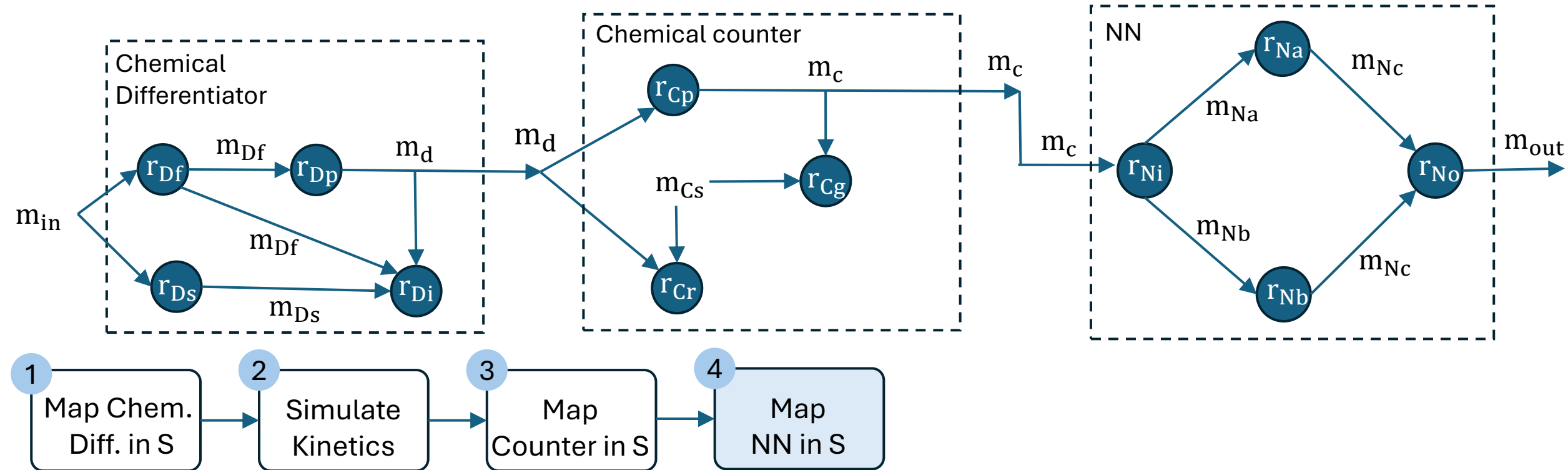
--- Candidate 6 ---

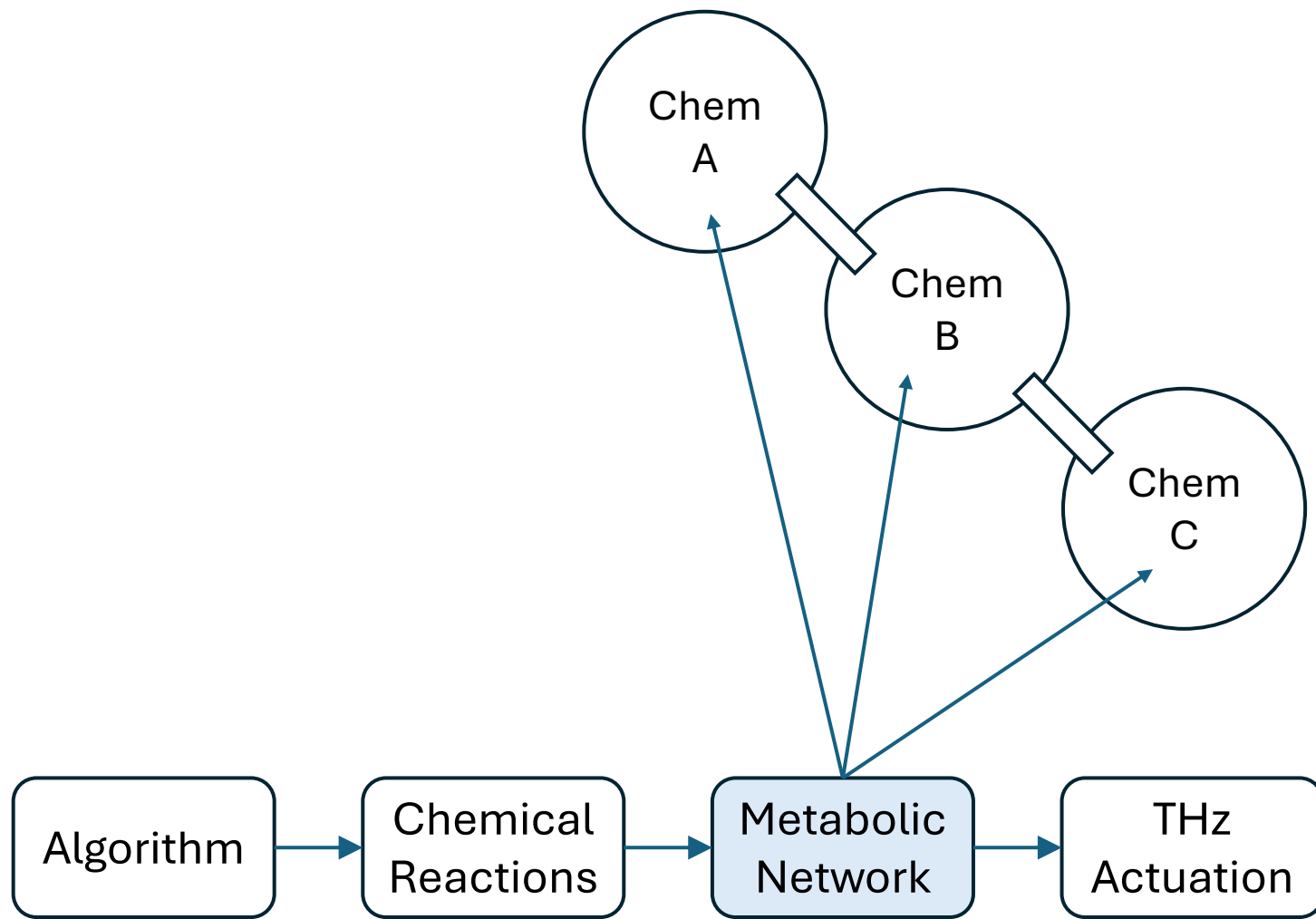
```

m_in  = accoa_c | Acetyl-CoA
m_Df  = occoa_c | Octanoyl-CoA (n-C8:0CoA)
m_d   = dcacoa_c | Decanoyl-CoA (n-C10:0CoA)
m_Ds  = malcoa_c | Malonyl-CoA
r_DF  = FAS80COA_L | Fatty acyl CoA synthase n C80CoA  lumped reaction
r_DS  = ACCOAC | Acetyl-CoA carboxylase
r_Dp  = FAS100COA | Fatty acyl CoA synthase n C100CoA
r_Di  = FAS120COA | Fatty acyl CoA synthase n C120CoA
GPR(r_DF) = YGR037C and YNR016C and (YKL182W and YPL231W)
GPR(r_DS) = YNR016C
GPR(r_Dp) = YGR037C and YNR016C and (YKL182W and YPL231W)
GPR(r_Di) = YGR037C and YNR016C and (YKL182W and YPL231W)

```







Enzyme unfolding probability

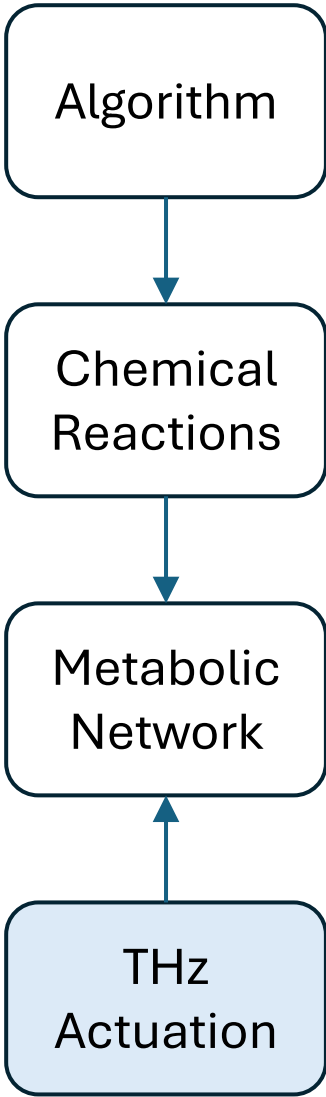
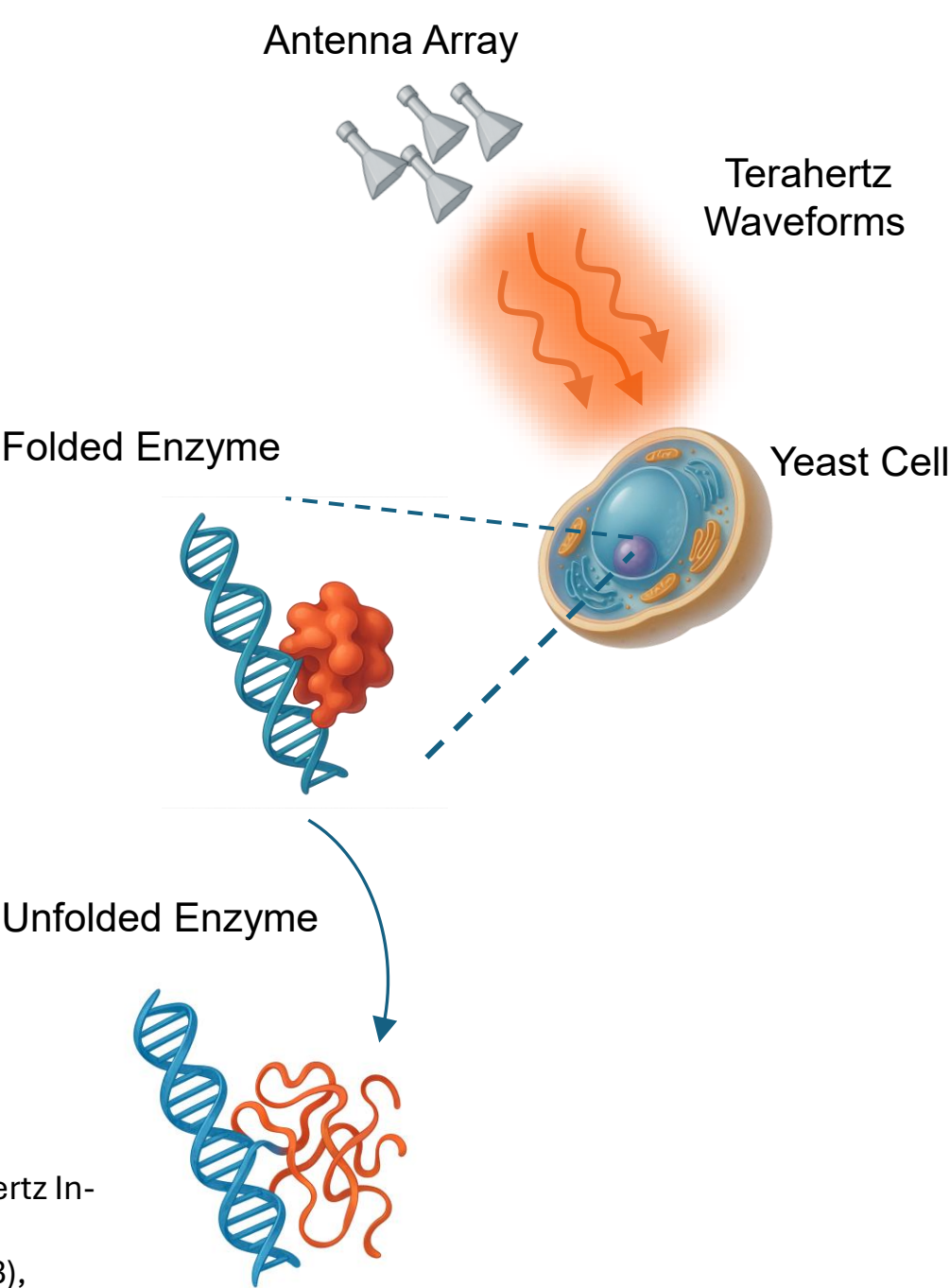
$$p_U = 1 - \frac{1}{1 + e^{\left(\frac{\Delta E - \langle E_{tot} \rangle_{ss}}{k_b T}\right)}}$$

$\langle E_{tot} \rangle_{ss}$ Total energy of the enzyme

ΔE Enzyme-free energy

k_b Boltzmann constant

T Temperature of the Cell

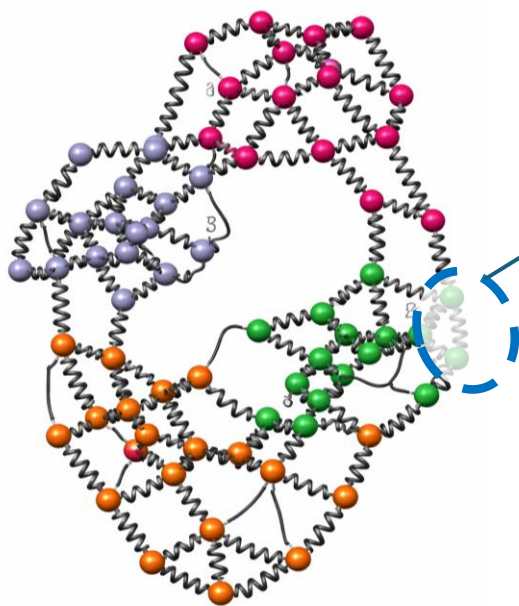


[1] Hadeel Elayan, Andrew W. Eckford, and Raviraj S. Adve. 2023. Terahertz Induced Protein Interactions in a Random Medium. IEEE Transactions on Molecular, Biological and Multi-Scale Communications 9, 4 (Dec. 2023), 435–446. <https://doi.org/10.1109/tmbmc.2023.3327302>

Enzyme
PBD ID: 1TUB



Anisotropic
Network
Model



Langevin Equation

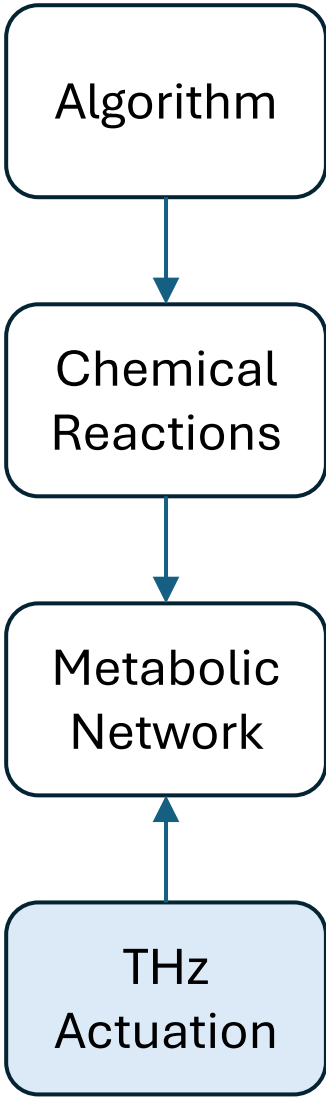
$$\mathbf{M}\ddot{\mathbf{x}} = -\mathbf{H}\mathbf{x} - \mathbf{B}\dot{\mathbf{x}} + \mathbf{f}_{\text{THz}}(t) + \boldsymbol{\xi}(t) \quad (1)$$

$$f_{\text{THz}}(t) = f_o \cos(\omega_a t + \Theta) \quad (2)$$

$$f_o = 2\pi r^3 \epsilon_m \text{Re}\{\mathcal{K}\} \nabla |\mathbf{E}_{\text{rms}}|^2 \quad (3)$$

$$p_U = 1 - \frac{1}{1 + e^{\left(\frac{\Delta E - \langle E_{\text{tot}} \rangle_{ss}}{k_b T}\right)}} \quad (4)$$

$$\langle E_{\text{tot}} \rangle = \frac{\alpha}{4\beta m [(\omega_o^2 - \omega_a^2)^2 + (\omega_a \gamma)^2]} (\omega_o^2 + \omega_a^2)^2 \quad (5)$$

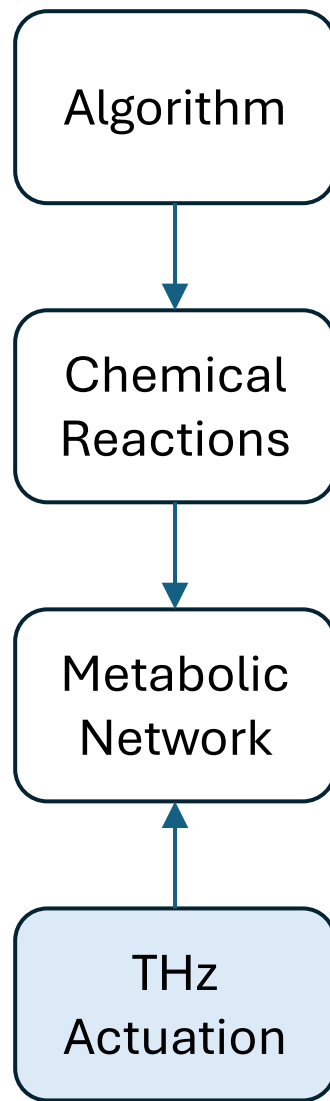
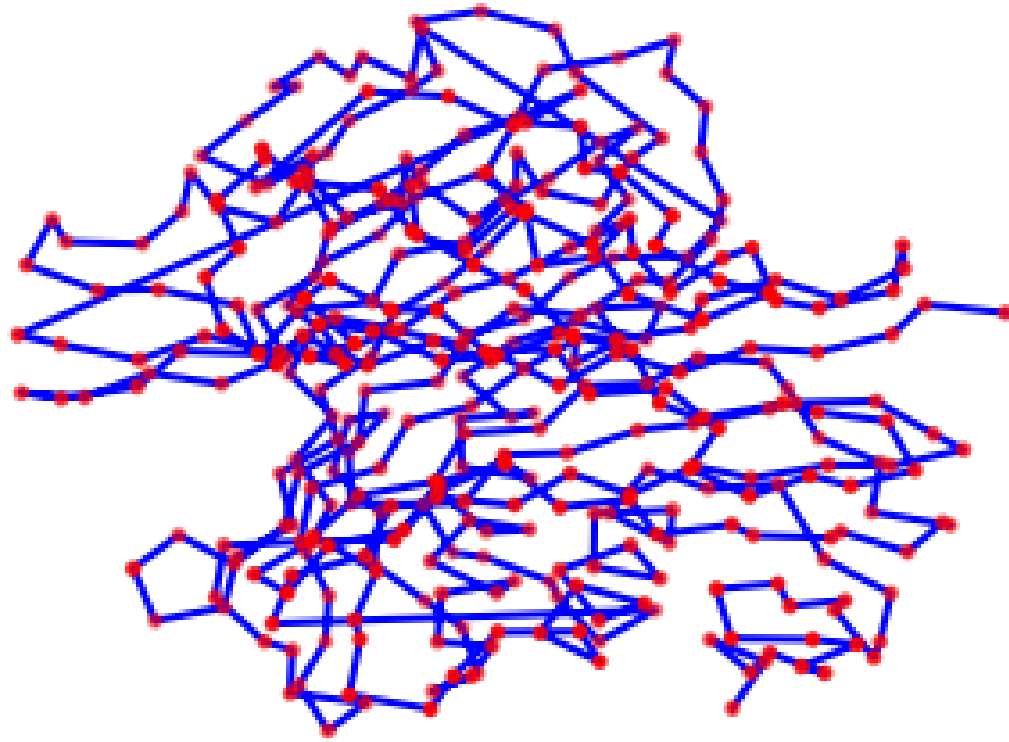


[1] Hadeel Elayan, Andrew W. Eckford, and Raviraj S. Adve. 2023. Terahertz Induced Protein Interactions in a Random Medium. IEEE Transactions on Molecular, Biological and Multi-Scale Communications 9, 4 (Dec. 2023), 435–446. <https://doi.org/10.1109/tmbmc.2023.3327302>

Enzyme PDB ID: 1TUB



Anisotropic Network Model (ANM)



Fundamentals of Flux Balance Analysis (FBA)

Stoichiometric matrix:

$$S_{\text{mets} \times \text{rxns}} = \begin{bmatrix} b_{11} & \cdots & b_{1,\text{rxn}} \\ \vdots & \ddots & \vdots \\ b_{\text{mets},1} & \cdots & b_{\text{mets},\text{rxns}} \end{bmatrix}, b_{ij} \in \mathbb{Z} \quad (3)$$

Reaction flux:

$$v \in \mathbb{R}^{|\text{rxns}|} \quad (4)$$

Example:

Reaction r_1 : $m_1 + 2m_2 \rightarrow m_3$

$$b_{11} = -1,$$

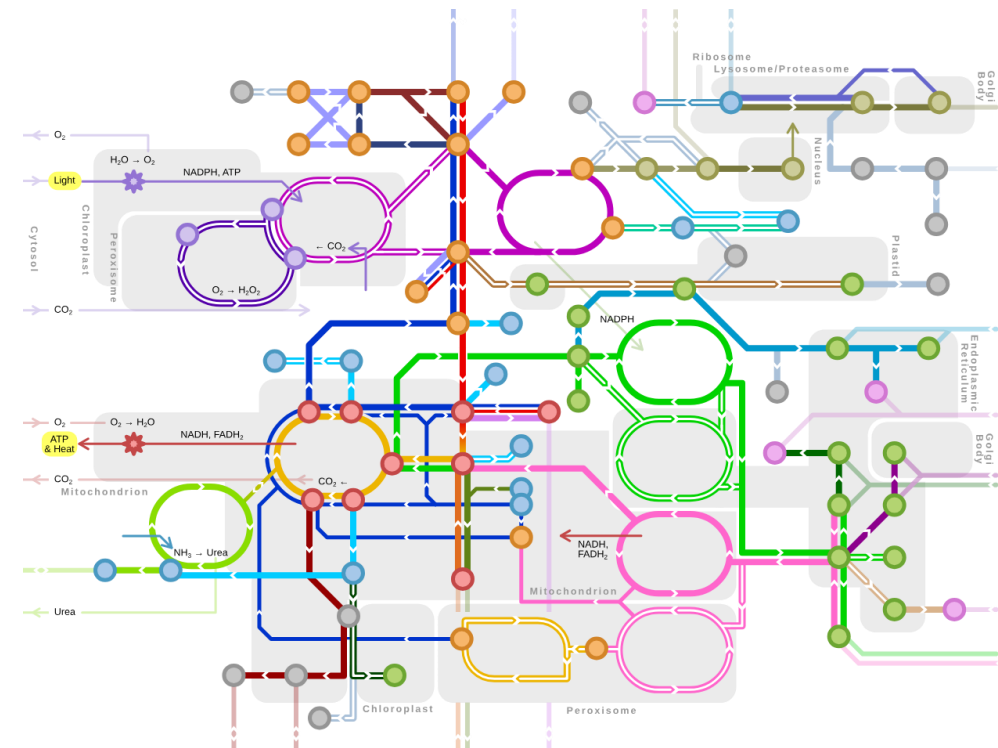
$$b_{21} = -2,$$

$$b_{31} = 1$$

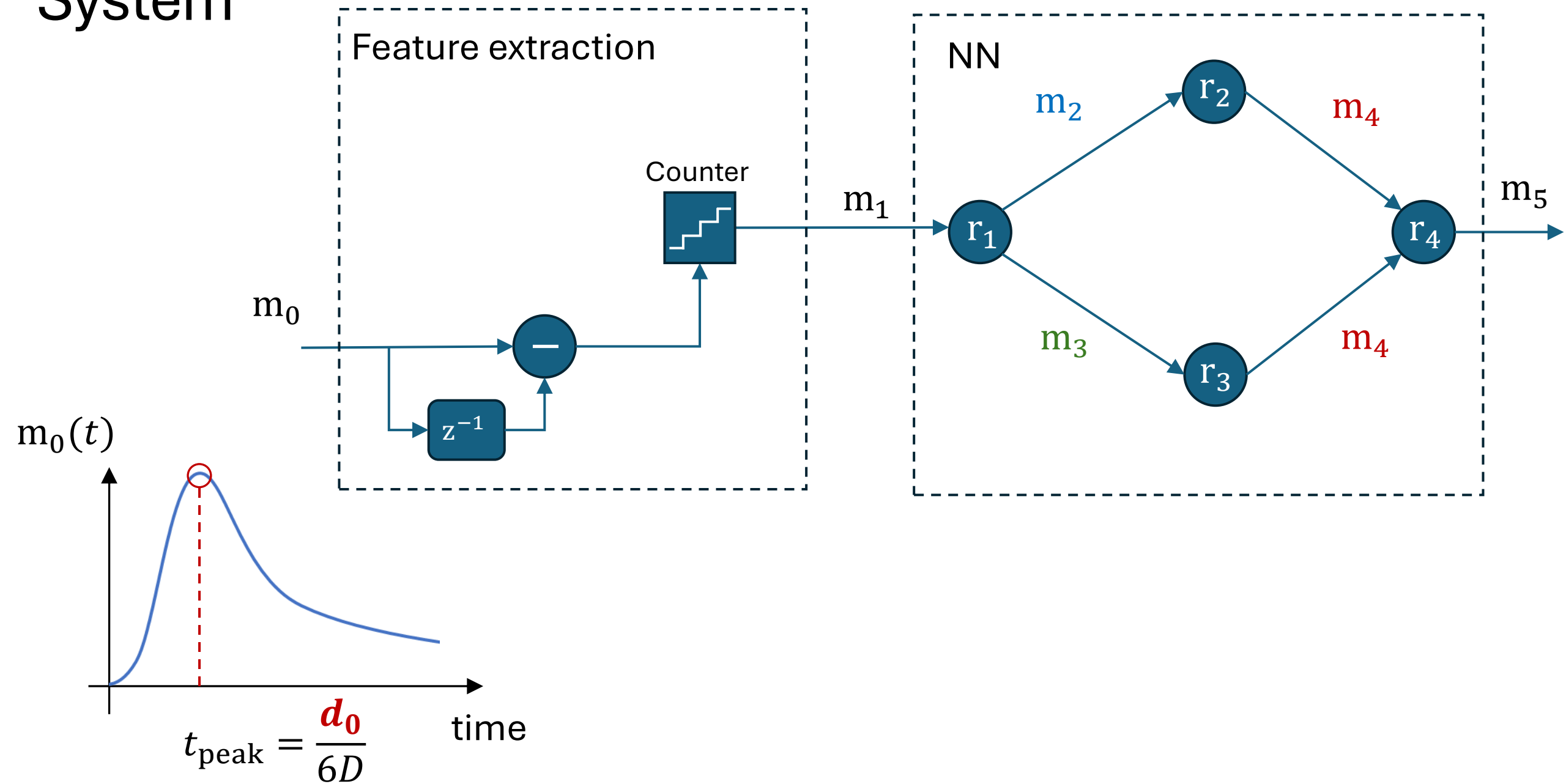
$$\frac{dx_m}{dt} = \sum_r S_{m,r} v_r = 0 \quad (1)$$

$\Downarrow$

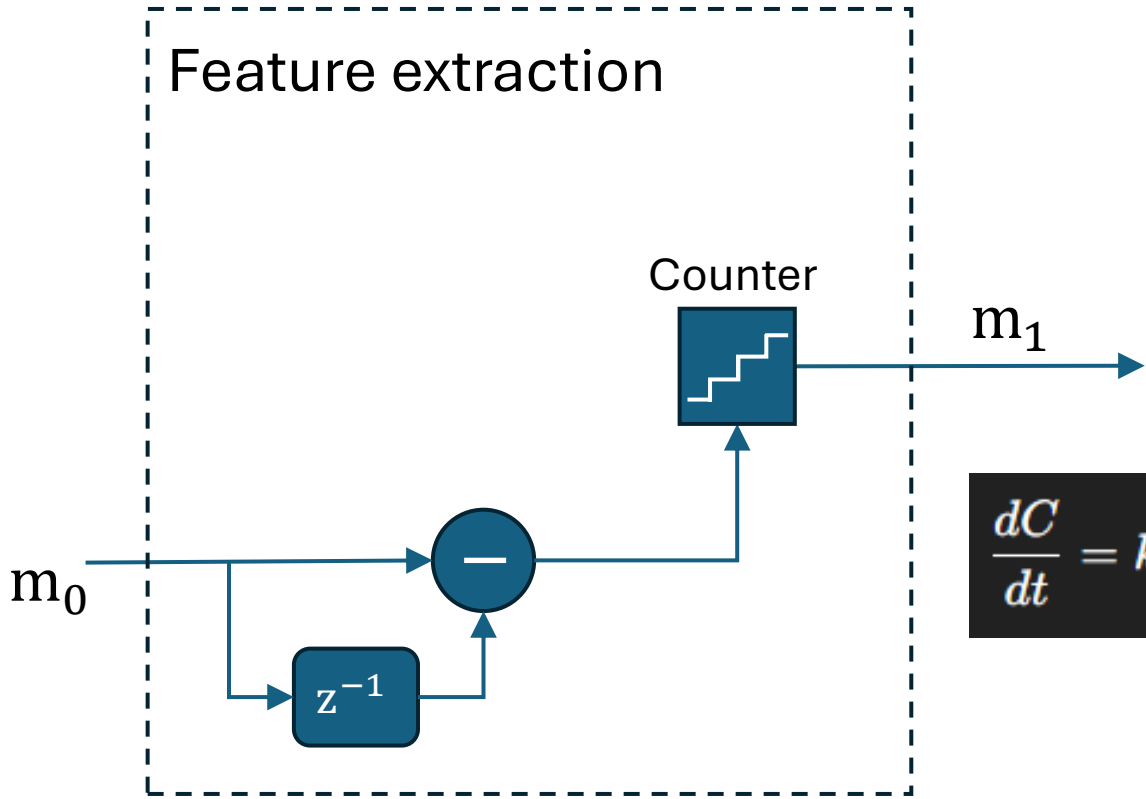
$$\begin{aligned} & \max_v c^T v \\ & \text{s.t. } Sv = 0 \\ & v \in [l \ u] \end{aligned} \quad (2)$$



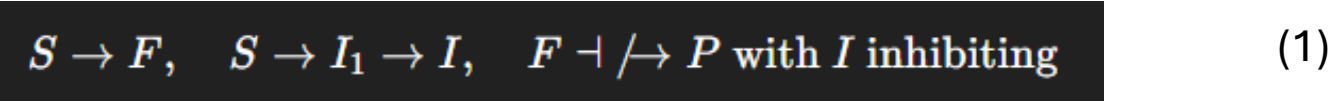
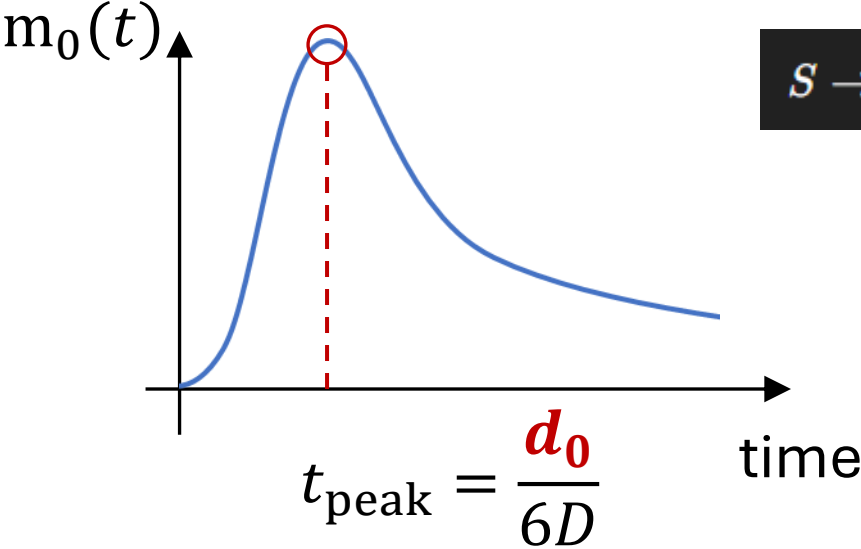
System

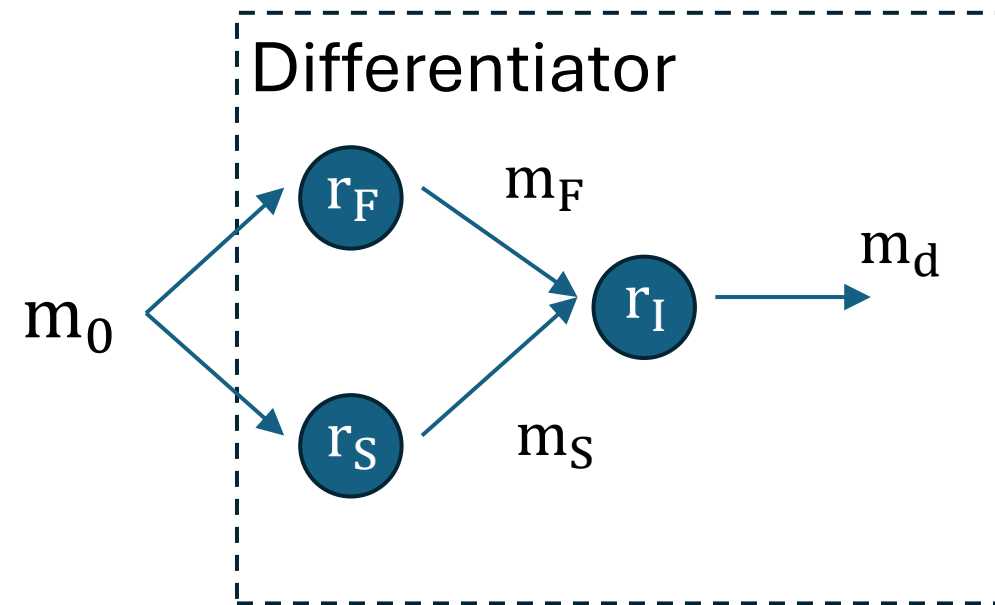
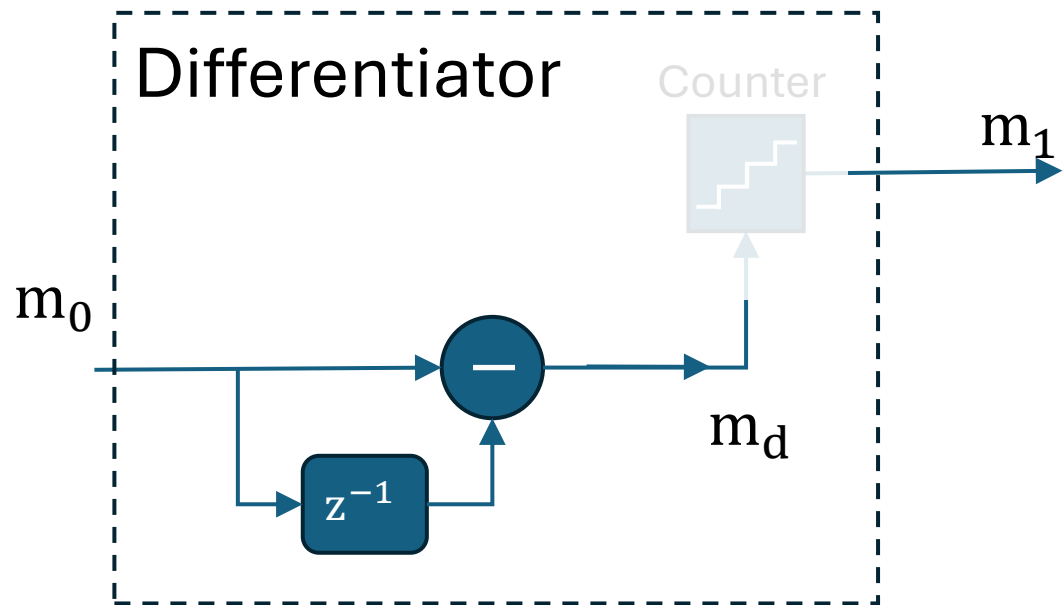


System



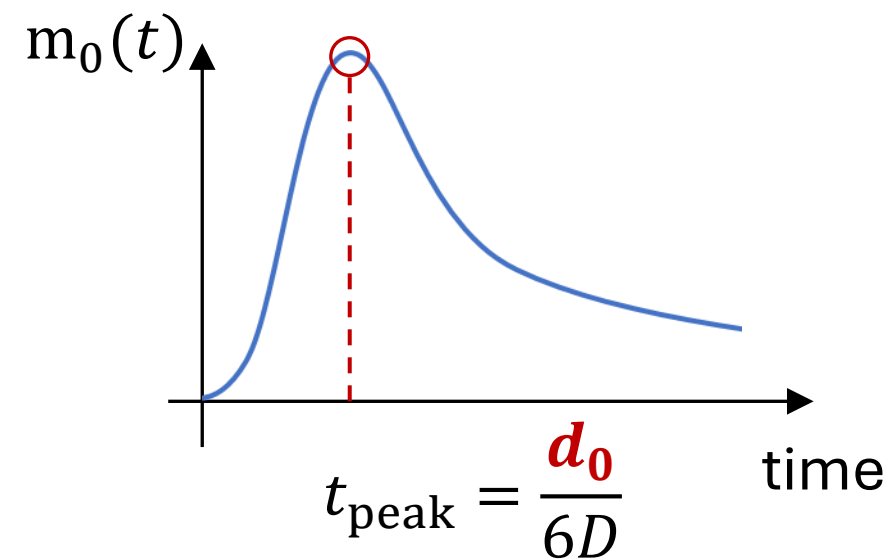
$$\frac{dC}{dt} = k_{\text{count}} \frac{P^n}{K_c^n + P^n} - k_{\text{reset}} \frac{K_r^m}{K_r^m + P^m} C \quad (2)$$

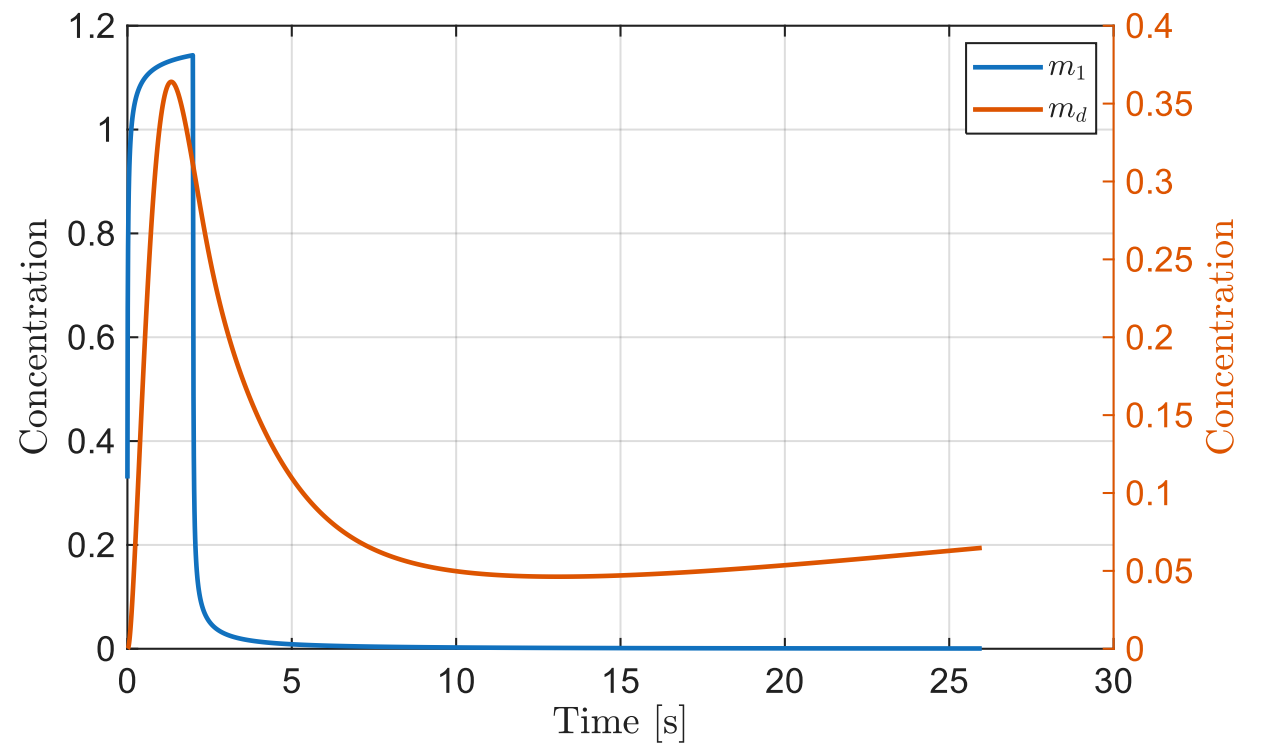
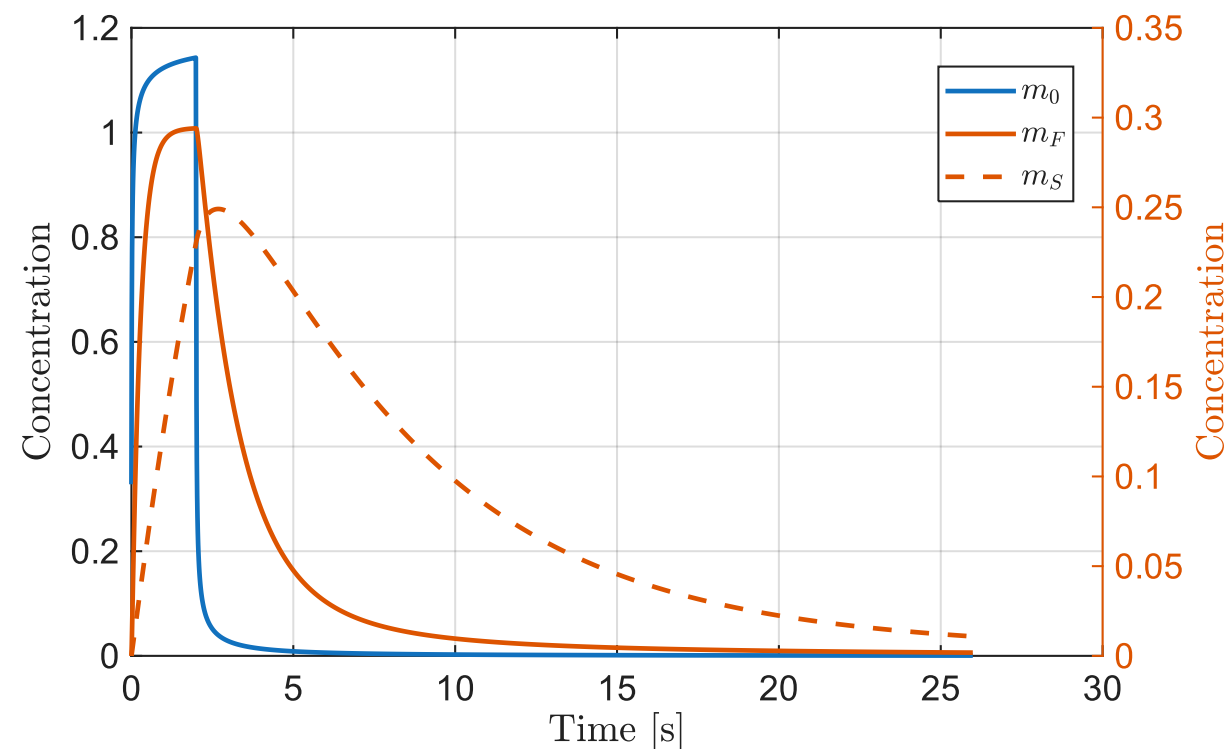
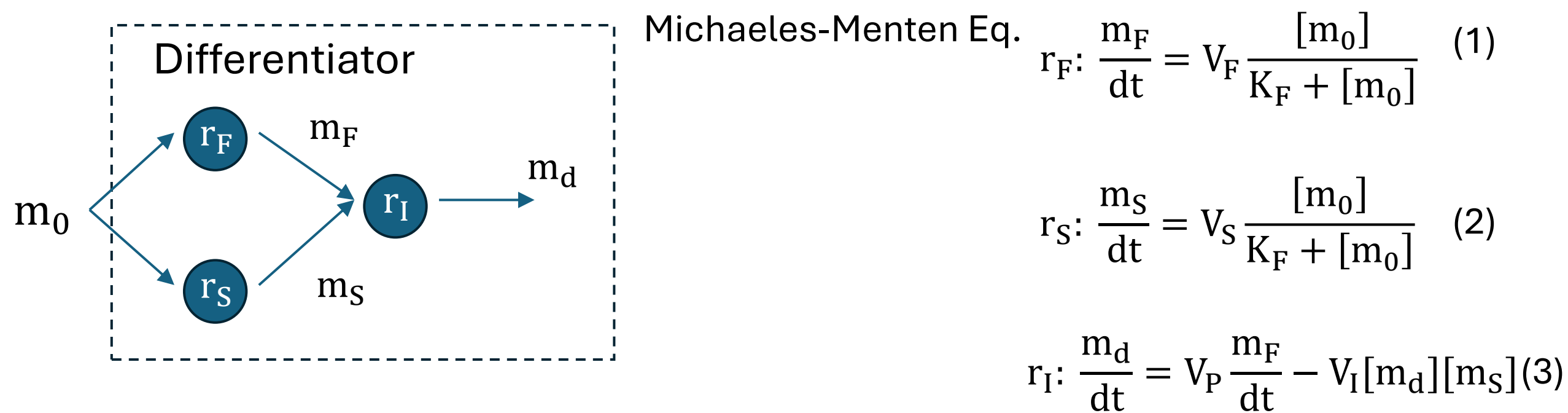




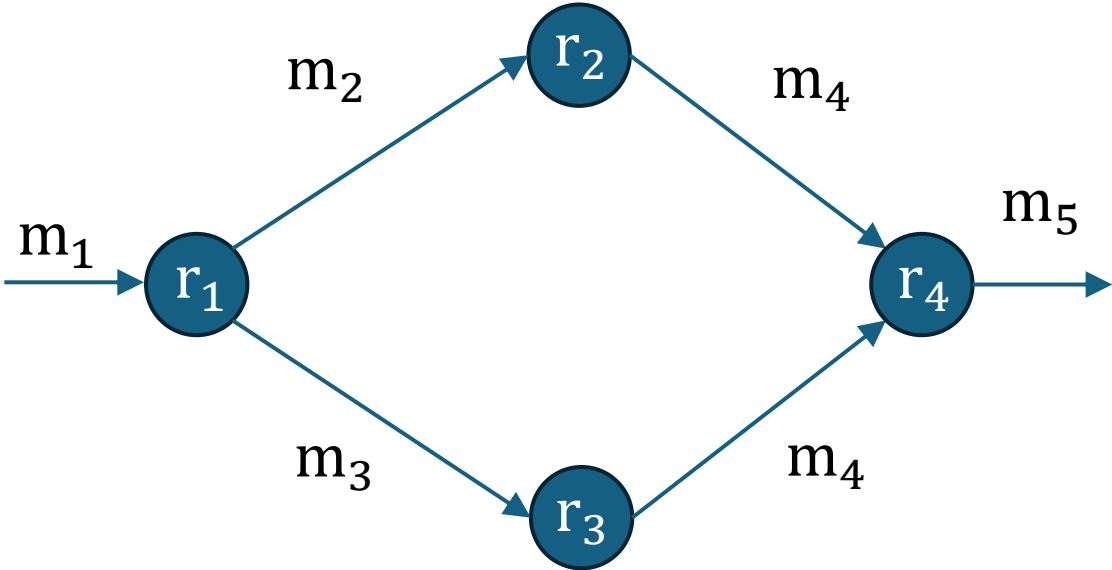
Point Transmitter

Receiver





Mapping a 2-neuron NN



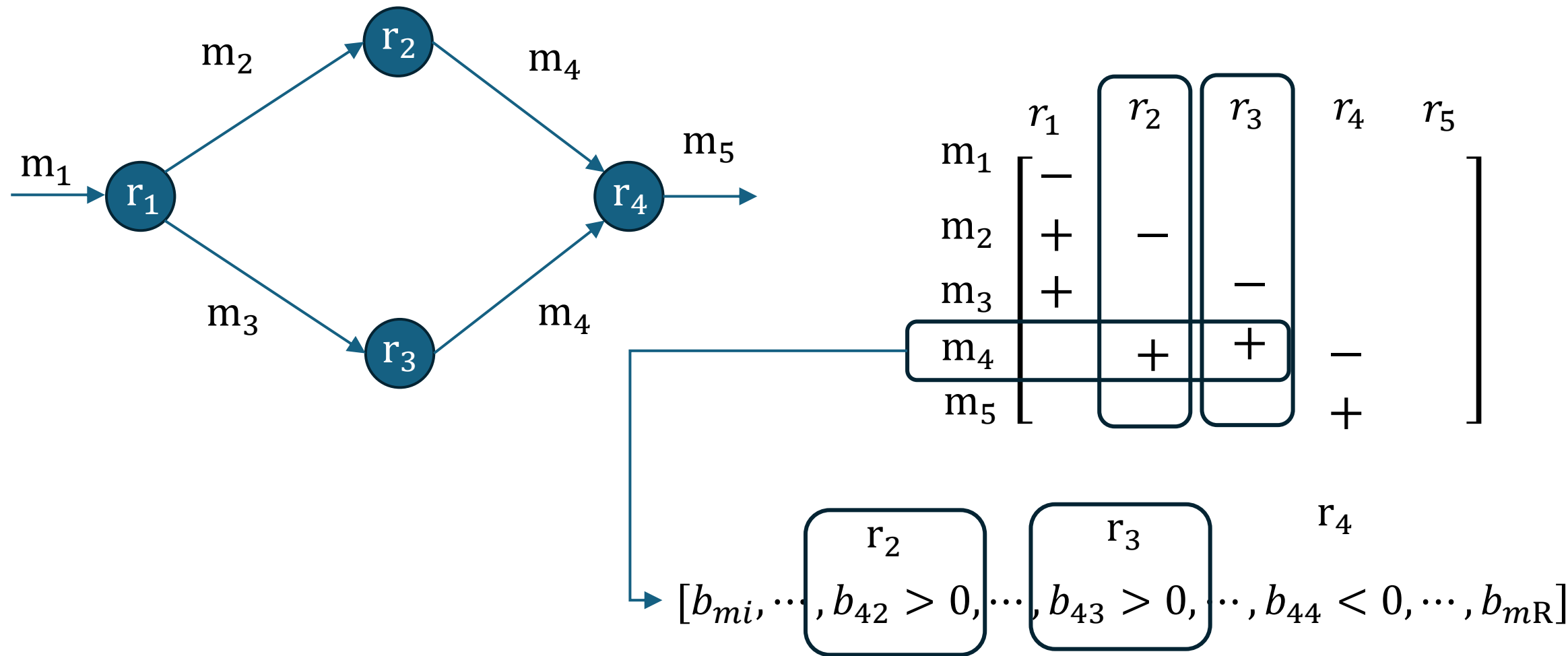
Pattern within the Stoichiometry Matrix

	r_1	r_2	r_3	r_4	r_5
m_1	-				
m_2	+	-			
m_3	+		-		
m_4		+	+	-	
m_5				+	

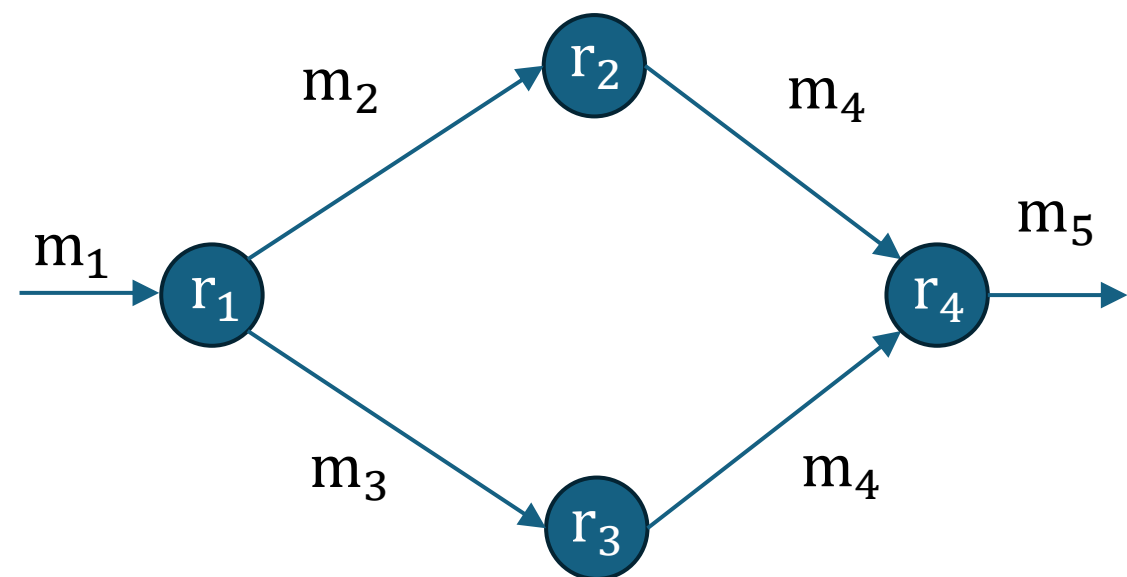
r_i r_j r_k

$[0, \cdots, b_{mi} > 0, \cdots, b_{mj} > 0, \cdots, b_{mk} < 0, \cdots, 0]$

Step 1: Find the m4-like metabolites and the reactions r₂'s and r₃'s

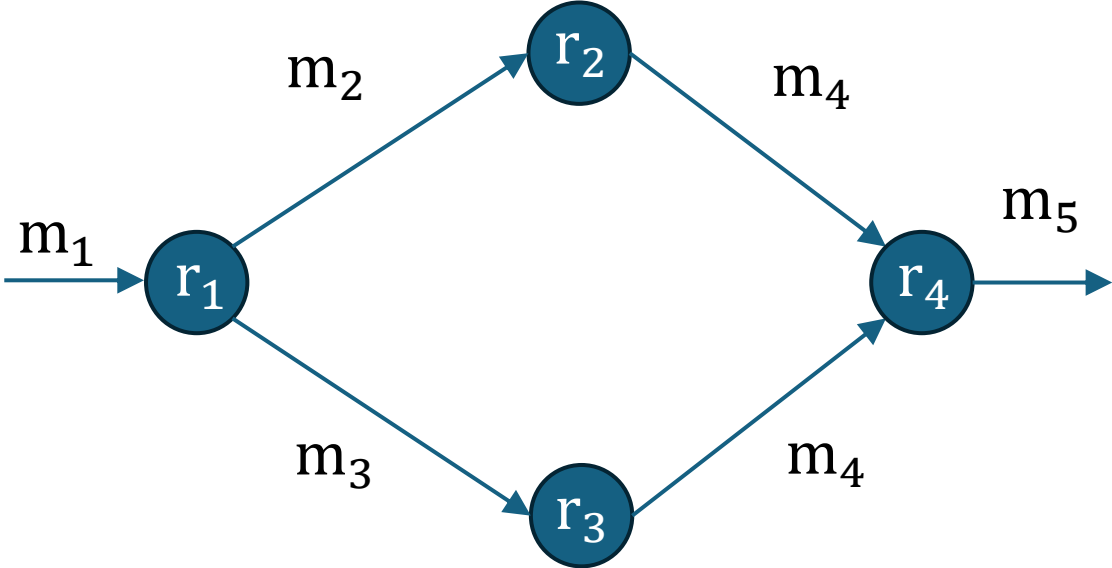


Step 1: Find the m4-like metabolites and the reactions r_2 's and r_3 's



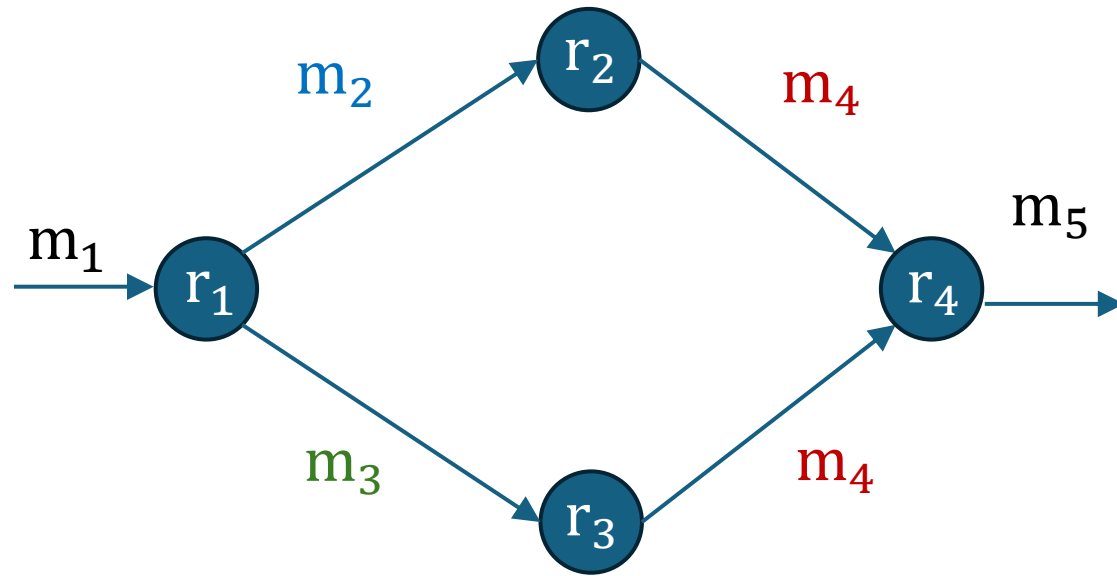
	m4	met	r2	r3	genes2	genes3	gpr2	gpr3
1	18	"s_0019"	591	592	"YMR208W"	"YMR208W"	"YMR208W"	"YMR208W"
2	18	"s_0019"	591	593	"YMR208W"	"YMR208W"	"YMR208W"	"YMR208W"
3	18	"s_0019"	591	594	"YMR208W"	"YMR208W"	"YMR208W"	"YMR208W"
4	18	"s_0019"	592	593	"YMR208W"	"YMR208W"	"YMR208W"	"YMR208W"
5	18	"s_0019"	592	594	"YMR208W"	"YMR208W"	"YMR208W"	"YMR208W"
6	18	"s_0019"	593	594	"YMR208W"	"YMR208W"	"YMR208W"	"YMR208W"
7	19	"s_0020"	3	78	"YAL060W"	["YGR087C"; "YLR...	"YAL060W"	"YGR087C or YL...
8	19	"s_0020"	3	731	"YAL060W"	["YGR087C"; "YLR...	"YAL060W"	"YGR087C or YL...

Step 2: Find reaction r_1 - like



	r_1	r_2	r_3	r_4	r_5
m_1	-				
m_2	+	-			
m_3	+		-		
m_4		+	+	-	
m_5				+	

Solutions



Metabolite m2: **C21H32N7O16P3S** | Name: coenzyme A| 405 | ID: s_0534

Metabolite m3: **C10H19O2** | Name: decanoate| 460 | ID: s_0600

Metabolite m4: **C31H50N7O17P3S**

Reaction 1: peroxisomal acyl-CoA thioesterase

EC code: 3.1.2.2

Gene: **YJR019C**

Formula:

$\text{C}_{31}\text{H}_{50}\text{N}_7\text{O}_{17}\text{P}_3\text{S} + \text{H}_2\text{O} \rightarrow \text{C}_{21}\text{H}_{32}\text{N}_7\text{O}_{16}\text{P}_3\text{S} + \text{C}_{10}\text{H}_{19}\text{O}_2 + \text{H}$

Location: Peroxisome

Reaction 2: acetyl-CoA C-acyltransferase (decanoyl-CoA)

EC code: 2.3.1.16

Gene: **YIL160C**

$\text{C}_{33}\text{H}_{52}\text{N}_7\text{O}_{18}\text{P}_3\text{S} + \text{C}_{21}\text{H}_{32}\text{N}_7\text{O}_{16}\text{P}_3\text{S} \rightarrow \text{C}_{23}\text{H}_{34}\text{N}_7\text{O}_{17}\text{P}_3\text{S} + \text{C}_{31}\text{H}_{50}\text{N}_7\text{O}_{17}\text{P}_3\text{S}$

Location: Peroxisome

Reaction 3: fatty-acid--CoA ligase (decanoate)

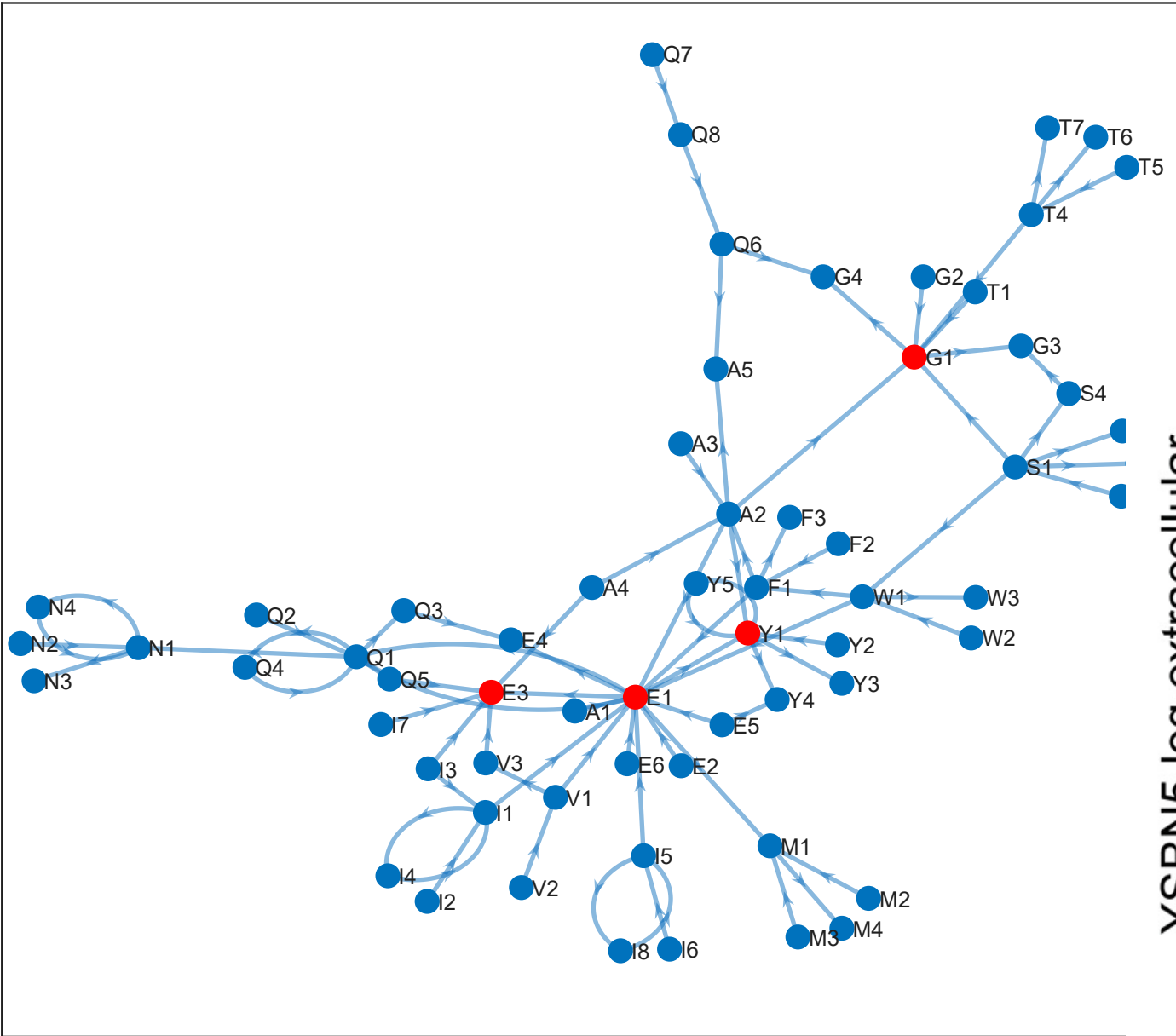
EC code: 6.2.1.3

Gene: **YER015W**

$\text{C}_{10}\text{H}_{12}\text{N}_5\text{O}_{13}\text{P}_3 + \text{C}_{21}\text{H}_{32}\text{N}_7\text{O}_{16}\text{P}_3\text{S} + \text{C}_{10}\text{H}_{19}\text{O}_2 \rightarrow \text{C}_{10}\text{H}_{12}\text{N}_5\text{O}_7\text{P} + \text{C}_{31}\text{H}_{50}\text{N}_7\text{O}_{17}\text{P}_3\text{S} + \text{HO}_7\text{P}_2$

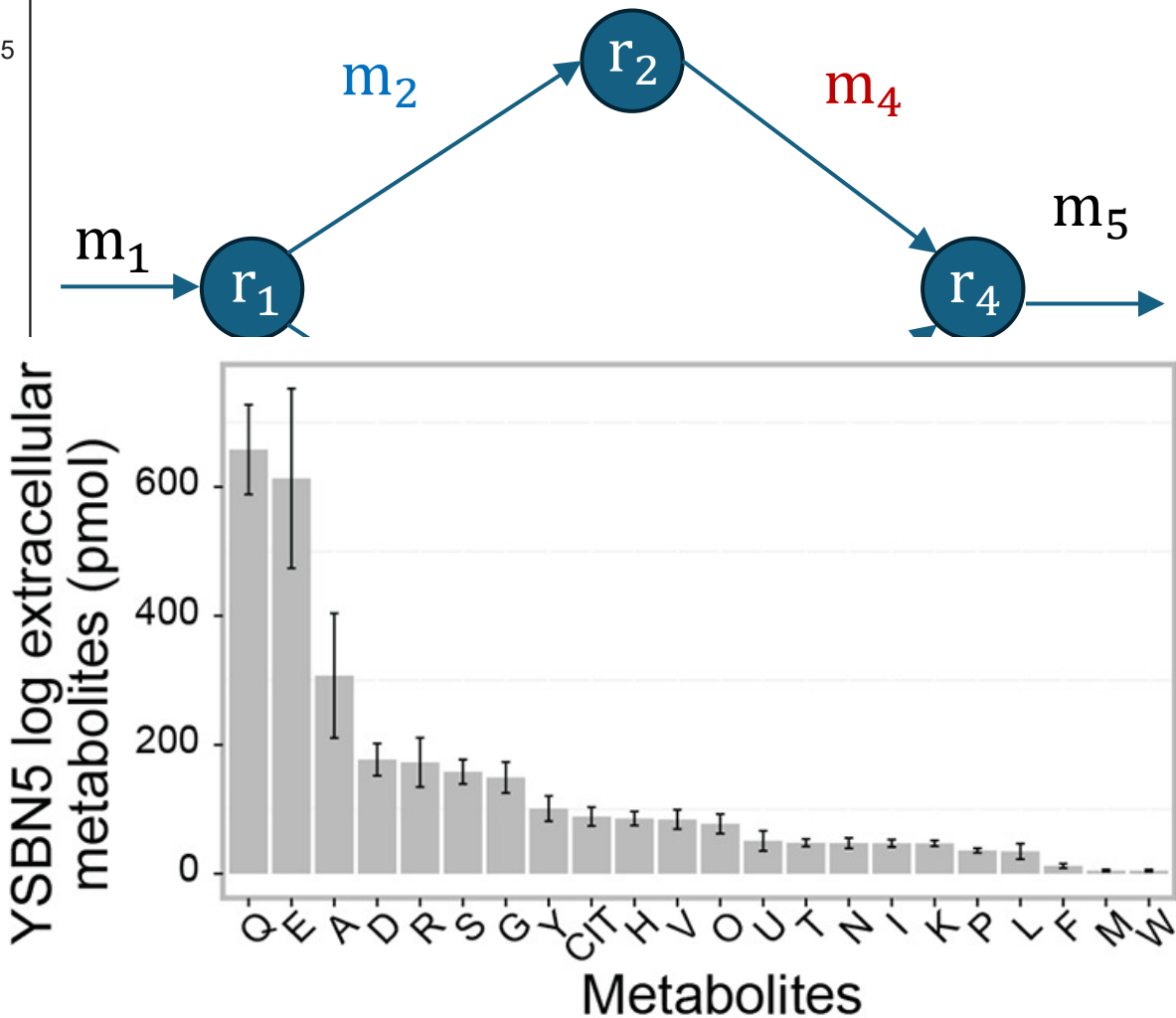
Location: Peroxisome

Mapping with restrictions

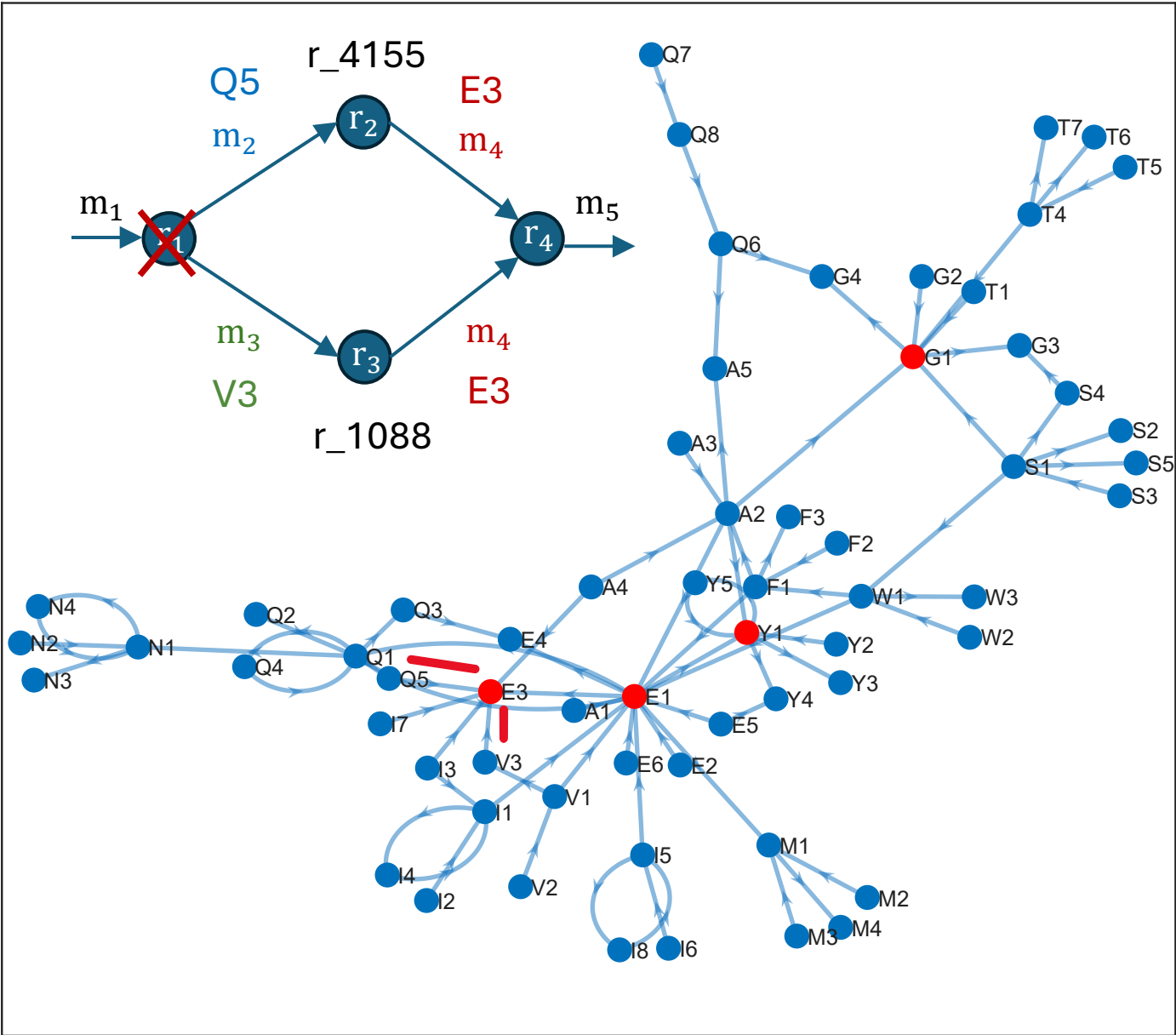


Restrictions:

1. The metabolite m_4 is a rich extracellular metabolite.
2. The metabolic network associated with m_4 is the least impactful.

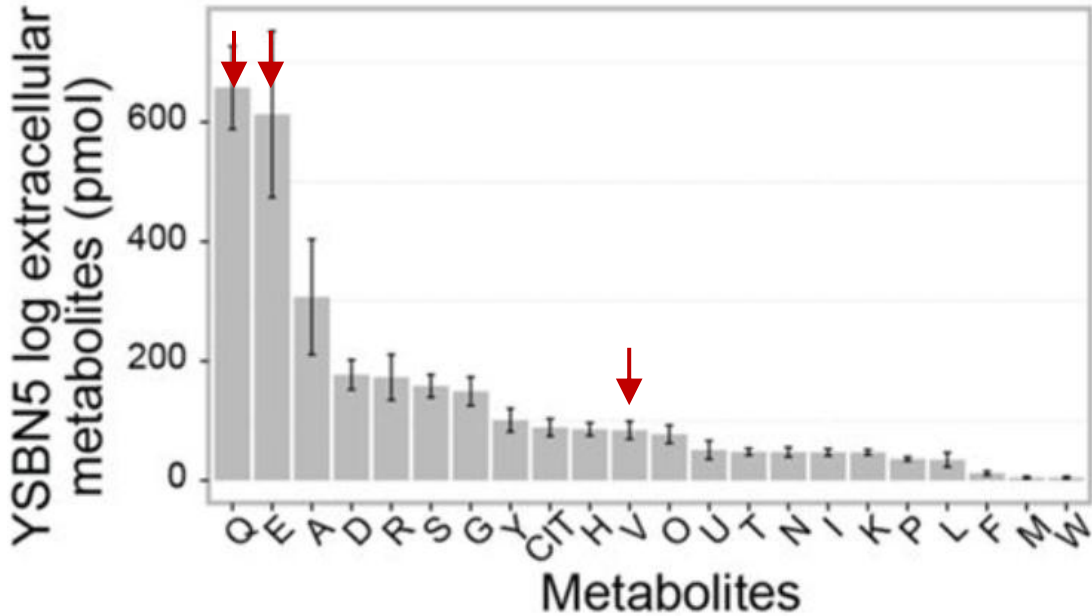


Mapping with restrictions



Restrictions:

1. The metabolite m_4 is a rich extracellular metabolite.
2. The metabolic network associated with m_4 is the least impactful.



Kate Campbell, Jakob Vowinckel, Michael Mülleder, Silke Malmshheimer, Nicola Lawrence, Enrica Calvani, Leonor Miller-Fleming, Mohammad T Alam, Stefan Christen, Markus A Keller, Markus Ralser (2015) Self-establishing communities enable cooperative metabolite exchange in a eukaryote eLife 4:e09943

THz-Induced Enzyme Unfolding Model

$$p_F = \frac{1}{1 + e^{\left(\frac{\Delta E - \langle E_{tot} \rangle_{ss}}{k_b T}\right)}} \quad (1)$$

$$\Delta E = E_f - E_u : \text{free energy} \quad (2)$$

$$\langle E_{tot} \rangle_{ss} = \frac{\alpha}{4\beta m [(\omega_o^2 - \omega_a^2)^2 + (\omega_a \bar{\gamma})^2]} (\omega_a^2 + \omega_o^2) + k_b T \quad (3)$$

$$\omega_o^2 = \frac{k}{m}, \omega_o \text{ is the natural frequency of the protein} \quad (4)$$

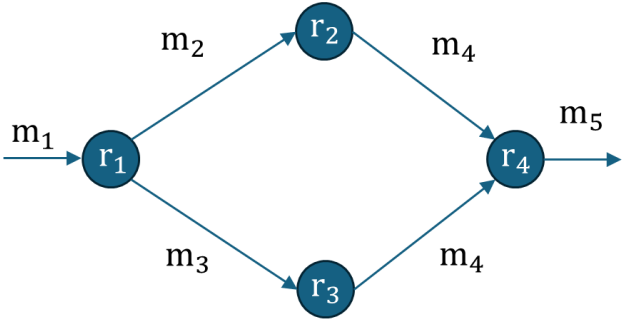
$$\bar{\gamma} = \frac{\ddot{\beta}}{m} \quad (5)$$

$$\ddot{\beta}: \text{Damping coefficient of the protein} \quad (6)$$

$$m: \text{Protein mass} \quad (7)$$

$$k: \text{Protein spring constant} \quad (8)$$

$$\omega_a: \text{Antenna radiation angular frequency} \quad (9)$$

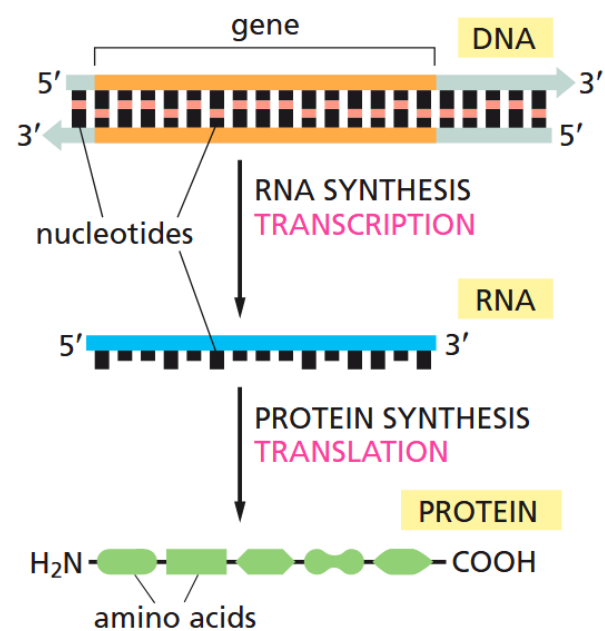


THI2

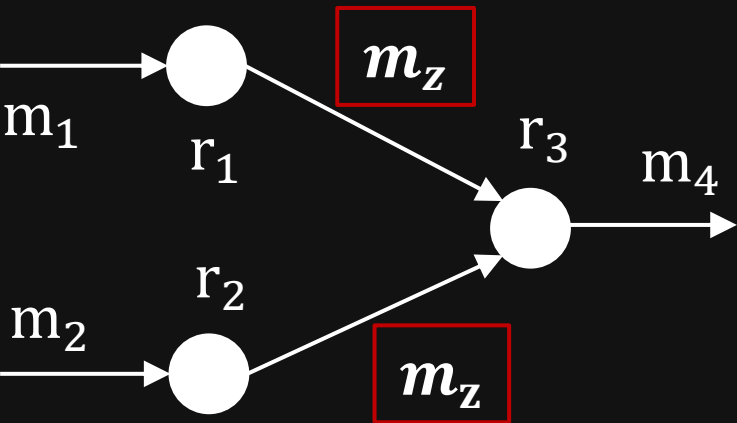
k: Protein spring constant

Physical Details	
Length (a.a):	450
Molecular Weight (Da):	51835.4
Isoelectric Point (pI):	8.63
Formula:	C ₂₂₈₅ H ₃₆₂₅ N ₆₃₅ O ₆₇₈ S ₃₁
Aliphatic Index:	76.96
Instability Index:	45.82

Looking for 2-neuron NNs



[1] B. Alberts, D. Bray, and K. Hopkin, Essential Cell Biology, 4th rev ed. edition. New York, NY: Norton & Company, 2014.



=== Motif 4 ===

Adder metabolite: s_{0293}

r1 (input): r_{0556}

$s_{0310} + s_{0434} \rightarrow s_{0293} + s_{0394} + s_{0794}$

$C_6H_9NOS + C_{10}H_{12}N_5O_{13}P_3 \rightarrow C_6H_8NO_4PS + C_{10}H_{12}N_5O_{10}P_2 + H$

gene: YPL214C

EC: 2.5.1.3;2.7.1.50

Location: cytoplasm

Subsystem: Thiamine metabolism

r2 (input): r_{4781}

$s_{0803} + s_{3910} \rightarrow s_{0293} + s_{0423} + s_{0456} + s_{0794}$

$H_2O + C_{17}H_{19}N_6O_{12}P_2S \rightarrow C_6H_8NO_4PS + C_{10}H_{12}N_5O_7P + CO_2 + H$

gene: YGR144W

EC: none annotated

Location: cytoplasm

Subsystem: Thiamine metabolism

z (output): r_{1036}

$s_{0267} + s_{0293} + s_{0794} \rightarrow s_{0633} + s_{1497}$

$C_6H_8N_3O_7P_2 + C_6H_8NO_4PS + H \rightarrow H_2O_7P_2 + C_{12}H_{16}N_4O_4PS$

gene: YPL214C

EC: 2.5.1.3;2.7.1.50

Location: cytoplasm

Subsystem: Thiamine metabolism